

Product Engineering Process (PEP), Directive 设计控制程序(PEP)

SSME CT QR 4.4/01

55 21 906 - AND - 308 - 11

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Restricted

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Content

0	History/ 修改历史.....	5
1	Purpose and scope / 目的和适用范围.....	7
2	Reference document / 参考文件.....	7
3	Abbreviations and Definitions/ 缩略语与定义.....	9
3.1	ABBREVIATIONS/ 缩略语.....	9
3.2	DEFINITIONS / 定义.....	11
3.2.1	Engineering sample/prototype / 工程样机 (EM).....	11
3.2.2	Drawing sample/prototype (ZM) / 图纸样机 (ZM).....	11
3.2.3	Production sample/prototype/ 生产样机 (FM).....	11
3.2.4	Formative evaluation/ 形成性评估.....	11
3.2.5	Summative evaluation/ 总结性评估.....	11
3.2.6	eSW/嵌入式软件.....	11
3.2.7	Intended Use/ 预期用途.....	12
3.2.8	Intended Purpose/ 预期目的.....	12
3.2.9	Software Component/ 软件组件.....	12
3.2.10	Non-Product Software/ 非产品软件.....	12
3.2.11	Component/组件.....	12
3.2.12	Hardware Component/硬件组件.....	12
4	Flow chart / 流程图.....	13
5	Process & Requirement/ 过程和需求.....	14
5.1	V-MODEL/REQUIREMENT TRACING/ V型模式/需求跟踪.....	14
5.2	STRUCTURE OF THE PRODUCT ENGINEERING PROCESS/ 产品设计过程结构.....	19
5.3	SYSTEM DEVELOPMENT PHASES AND THEIR PURPOSE/ 系统开发阶段及其目的.....	20
5.3.1	Process Phase 1: Idea Collection / 过程阶段1: 想法收集.....	21
5.3.2	Process Phase 2: Product Definition / 过程阶段2: 产品定义.....	21
5.3.3	Process Phase 3: Product Development / 过程阶段3: 产品开发.....	25
5.3.4	Process Phase 4: Product Support / 过程阶段4: 产品支持.....	29
6	General requirements/ 通用要求.....	32
6.1	DESIGN AND DEVELOPMENT PLANNING/ 设计和开发计划.....	32
6.2	TAILORING/ 裁减.....	32
6.2.1	Project Specific Tailoring/ 项目特定裁减.....	32
6.2.2	Software Specific Tailoring/ 软件特定裁减.....	32
6.2.3	Standard Tailoring for Successor Versions of Platform Release/ 平台发布后续版本的标准裁减.....	32
6.2.4	Standard Tailoring for Agile Approaches during Product Development/ 产品开发期间敏捷方法的标准裁减.....	32
6.3	REQUIREMENTS FOR THE PRODUCT/ 产品需求.....	33
6.4	FURTHER REQUIREMENTS FOR DEVELOPMENT AND MAINTENANCE/开发和维护的进一步要求.....	33
6.4.2	Claims/ 声明.....	33
6.4.3	Product Security/ 产品网络安全.....	33

6.4.4	AI Systems/ 人工智能系统	33
6.5	REQUIREMENT TRACING/ 需求追踪	34
6.6	AUTOMATED PROCESSES/ 自动过程	34
6.7	RISK MANAGEMENT/ 风险管理	34
6.8	PRODUCT RELIABILITY/产品可靠性	34
6.9	PRODUCT-RELATED ENVIRONMENTAL PROTECTION	34
6.10	STANDARDS & LAWS/ 标准和法律	35
6.11	PATENTS AND TRADEMARKS/ 专利和商标	35
6.12	PROCUREMENT AND USE OF THIRD-PARTY SOFTWARE/ 采购和第三方软件使用	35
6.13	PREVENTIVE ACTION/ 预防措施	35
6.14	CONCURRENT ENGINEERING/ 并行开发	36
6.15	TRANSFER TO PRODUCTION/ 生产转移	36
6.16	CHANGE CONTROL PROCEDURE/修改控制程序	36
6.16.1	Changes during product development/ 产品开发中的更改	36
6.16.2	Changes after Release /Product enhancements/ 发布后的变更/产品增强	36
6.17	DESIGN REVIEW / TECHNICAL REVIEW/ 设计评审/技术评审	36
6.17.1	Design review/ 设计评审	36
6.17.2	Technical review / Content review/ 技术评审/内容评审	37
6.18	PROJECT PHASE APPLICATION/ 项目阶段申请	37
6.19	HANDLING DEVIATIONS/偏差处理	37
6.20	DESIGN VERIFICATION/ 设计验证	38
6.21	DESIGN VALIDATION/ 设计确认	38
6.22	PROCESS VALIDATION/ 过程确认	40
6.23	CLINICAL EVALUATION/ 临床评估	40
6.24	SOFTWARE DEVELOPMENT/软件开发	40
6.25	OTS SOFTWARE/ 现成软件	40
6.26	UNIQUE DEVICE IDENTIFICATION/ 唯一的设备识别	41
7	Document/ 文档	41
7.1	DOCUMENT MANAGEMENT/ 文档管理	41
7.2	QUALITY MANAGEMENT PLAN (QMP)/ 质量管理计划 (QMP)	41
7.3	PROJECT MANAGEMENT PLAN (PMP) / 项目管理计划 (PMP)	42
7.4	PRODUCT REQUIREMENTS SPECIFICATION LEVEL/ 产品需求规范层级	43
7.5	SYSTEM REQUIREMENTS SPECIFICATION LEVEL/ 系统需求说明级别	43
7.6	SUBSYSTEM / COMPONENT LEVEL/ 子系统/部件级别	45
7.7	SYSTEM REALIZATION CONCEPT/ 系统实施概念	46
7.8	TEST CONCEPT/ 测试概念	46
7.9	RESULTS FROM RISK MANAGEMENT/ 来自于风险管理的结果	46
7.10	ERROR PREVENTION REPORT/错误预防报告	46
7.11	DOCUMENT PLAN / DESIGN HISTORY FILE (DHF) / 文档计划/设计历史文件 (DHF)	47
7.12	DEVICE HISTORY RECORD (DHR) / 产品追溯性记录 (DHR)	47
7.13	DEVICE MASTER RECORD (DMR)/ 产品制造性文档 (DMR)	48
7.14	MEDICAL DEVICE FILE (MDF)/ 医疗器械文档 (MDF)	48
7.15	TECHNICAL DOCUMENTATION/ 技术文档	49
7.16	COUNTRY-SPECIFIC APPROVALS (REGULATORY APPROVAL PLAN) / 特定国家核准 (法规核准计	

划)		49
7.17	AFFIDAVIT FOR SERVICE SOFTWARE/ 服务软件宣誓书	49
7.18	USABILITY ENGINEERING FILE/ 可用性工程文件	49
7.19	SYSTEM FLOWCHART/ 系统流程图	50
7.20	ACCOMPANYING DOCUMENTS/ 随机文档.....	50
7.21	FINANCIAL PRODUCT PLAN (FPP) / 财务产品计划	50
8	Tasks and responsibilities/ 任务和职责	50
8.1	MANAGEMENT OF SSME CT / SSME CT管理层	50
8.2	QUALITY MANAGEMENT REPRESENTATIVE / 质量管理负责人	51
8.3	PRODUCT MANAGER / 产品经理	51
8.4	PROJECT MANAGER / 项目经理	51
8.5	PRODUCT QUALITY MANAGER / 产品质量经理	52
8.6	PRODUCT STEERING GROUP (PSG)/ 项目管理小组成员	53
8.7	RESPONSIBILITIES DURING DELIVERY / 发运中的职责.....	53
9	Provisional solution and backward method/ 过渡措施和补救办法.....	54
10	Record/记录	54
11	Appendix/附录	54

0 History/ 修改历史

Nr.	Page	Version	Change description	CR No.
1	All	01	Initial creation, the content transferred from 55 21 906 – AND – 008 – 007 due to SSME QMS restructuring.	N.A.
2	All	02	Change content: Alignment w/ process manual 4791500-AND-105 to cover both Emotion and aCTivate product development.	P10222
3	All	03	Main changes: combine system and software projects milestones; shift R3 before product verification; other adaptations like renamings MEP→CUT, V&V descriptions and requirement levels rework, etc.	P10351
4	All	04	Change content: Adapt MEDDEV 2.7.1 Revision 4; Adapt ISO 13485: 2016; Reference document name update. Add UDI requirement Add software development requirement. Update the requirement for CT QR4 4791500-AND-105-17.	P10371, P10380, P10414, P10440,
5	All	05	E-QMS Project	P10626
6	All	06	Update due to QR28-Product Reliability Management	PCR10803
7	All	07	Chang Reason: 1. supplier management procedure change. and QR28 was integrated into QR0 2. CT20210014-for wrong reference of “the supplier Quality management procedure”. Chang Content: 1.Change the reference from 5521906-AND-304 to 4791600-QMS-700 “SOP - DI Supplier Quality Management 2.Change reference from 11106079-AND-02S QR28-Product Reliability Management to 11106101-AND-02S QR0-Basic Quality Requirements for Business Areas/Lines and Technology Excellence Units Appendix 2 roles updated, reference doc. updated 5521906-AND-269 change to 5521906-AND-376 5521906-AND-285 change to 5521906-AND-378	PCR 10872
8	All	08	Change Reason: - add reference to Software development 5521922 AND 320 - alignment with FO PEP 4791500-AND-105-19 Change Description: - add reference to Software development 5521922 AND 320 in related chapters - add definition of intended use and intended purpose in chapter 3.2 - add definition of software component, software development plan in chapter 3.2 - add tailoring process in chapter 6.2 - add claims in chapter 6.4 - add OTS software in chapter 6.25 - add abbreviations in chapter 3.1 - other contents alignment with FO PEP	PCR 11002

			4791500-AND-105-19	
9	All	09	Change Reason - update PEP to include CT WI 4.4/03 and CT WI 4.4/19 - To include process on Early Procurement Involvement (EPI) and Early Supplier Involvement (ESI) Change Description: -Add CT WI 4.4/19 5521922-AND-319 into references and chapter 3.2.10 software Development Plan -Add CT WI 4.4/03 5521922-AND-302 into references and chapter 7.15 Country-specific approvals -Add EPI and ESI related description in development phase before M200	PCR 11095
10	All	10	Change reason: - Get alignment with FO PEP 4791500-AND-105-20, changes for understandability and consistency with project practice. Change Description: - Chapter 3.1, updated/new added abbreviations: removed D1-D5, EK; new added PMA(CR240015); adapted R1-R5, pR xx. - Chapter 3.2, updated/new added definitions: adapted contents with yellow shading, removed "software development plan" and new added "component" and "hardware component" (CR241286) and "non-product software". - Chapter 5.1, updated figure 1 of V model and relevant description (CR241286). - Chapter 5.2, updated figure 2 of Structure of the Product Engineering Process and relevant description (CR241286). - Chapter 6, 7 and 8, contents modification with yellow shading (CR241286/CR240567/CR220799/CR240745).	PCR 11314
11	All	11	Change Reason: 1. Update PEP to change QSR to QMSR 2. Update requirements from (EU 2024/1689) AI Act gap analysis 3. CAPA CT20250002 PA1 was assessed that No report template is referenced, therefore no action is required in PEP appendix 4. Typo and translation error correction Change Description: 1. Change all QSR to QMSR, updated all the related naming of 21CFR820 subpart by removing "FDA QSR". 2. Chapter 3.1, remove QSR and new added QMSR definition. For terms and definitions of DHF, DMR & DHR which were removed from new QMSR, these terms/definitions are retained in SSME CT QMS as its historical usage, familiarity and other regulatory definitions/clauses. 3. Chapter 3.1 Make the definition of EOD more explicit. 4. Chapter 5.3.2, 5.3.3 new added AI related activities. 5. Chapter 5.3.3, in phase of R3-R4, change validation to verification. 6. Chapter 5.3.4, change (EOS) to (EOD)	PCR 11421

			7. Chapter 6.4.4 new added AI systems	
			8. Appendix 2 translation correction	

1 Purpose and scope / 目的和适用范围

This CT QR shall be effective for SSME CT where the QMM [6] is applicable.

It regulates the procedure for planning, R&D, management and further development of systems, sub-systems, system components and services and purchases included therein. It describes the general guidelines and typical requirements, which must be observed for all R&D projects.

Depending on the particular project circumstances, these data can be adapted to requirements, as long as this does not run counter to the basic principle of the CT QR (tailoring is to be described in the QMP)

For technical advancements, for example, the formal M120 and M150 decisions may be omitted if all the required design input documents are present.

本程序适用于在质量管理手册[6]适用范围内的 SSME CT。

本程序规定了对系统、子系统、系统组件进行计划、技术开发及管理/进一步开发的程序，维修和采购也包含在内。本指令还对所有技术开发项目必须遵守的总方针以及典型需求作了说明。

只要不违背质量保证程序基本原则，本规定可以根据项目的特殊情况进行修改以适合具体的需求(调整和裁剪 – 在质量管理计划中描述)。

比如，对于进一步技术开发，只要提交所有所需的设计输入文件，可以省略正式 M120 和 M150 决定。

2 Reference document / 参考文件

[1]	5521906-AND-305	Change Control Procedure	修改控制程序
[2]	4791600-QMS-700	Standard Operating Procedure Supplier Quality Management	供应商质量管理 标准操作程序
[3]	8596541-AND-107	Design Transfer Guidance	设计转移指导书
[4]	5521906-AND-319	Process Validation and Verification Procedure	过程确认与验证程序
[5]	5521906-AND-348	Customer Complaint Handling Procedure	用户抱怨处理程序
[6]	5521906-AND-300	Quality Management Manual (QMM)	质量管理手册
[7]	5521906-AND-210	Document Management Procedure	文件管理程序
[8]	5521906-AND-352	Product Risk management Procedure	产品风险管理程序
[9]	5521906-AND-302	Management responsibility	管理职责
[10]	5521922-AND-360	Product Phase-out Process at Product Lifecycle Management Work Instruction	产品生命周期管理中的产品逐步退出流程作业指导书
[11]	5521922-AND-355	Field test for Customer Acceptance and Usability in the Clinical Use Test (CUT) Clinical Investigation Work Instruction	CUT 中用户接受度和可用性 现场测试, 临床调研作业指导书
[12]	5521922-AND-323	Customer Documentation Work Instruction	用户文档管理作业指导书
[13]	8596541-AND-008	Milestone Review Checklist	项目结点确认表

[14]	8596541-AND-1RD	Reference Data for CT-Documents	CT 文档参考表
[15]	5521922-AND-348	Device History Content Requirements Work Instruction	产品追溯记录作业指导书
[16]	5521906-AND-270	Software that is not part of medical device	非医疗设备的软件
[17]	5521906-AND-376	Regulatory Requirement	法规需求
[18]	5521922-AND-368	Performing Review Instruction	评审指导书
[19]	5521906-AND-374	Product Usability Management Procedure – Safety Aspects of Usability	产品可用性管理流程 – 安全相关的可用性
[20]	5521906-AND-378	Clinical Evaluation in European Union	欧盟临床评估
[21]	5521906-AND-345	Choosing and Applying the Statistical Technology Procedure	统计技术的选定和应用程序
[22]	21CFR820.3(f)	Definitions: Design Input	定义: 设计输入
[23]	21CFR820.30(c)	Design Controls: Design Input	设计控制: 设计输入
[24]	21CFR820.3(g)	Definitions: Design Output	定义:设计输出
[25]	21CFR820.30(d)	Design Controls: Design Output	设计控制:设计输出
[26]	21CFR820.30(h)	Design Controls: Design Transfer	设计控制:设计转移
[27]	21CFR820.3(z)	Definitions: Design Validation	定义:设计确认
[28]	21CFR820.30(g)	Design Controls: Design Validation	设计控制:设计确认
[29]	21CFR820.3(aa)	Definitions: Design Verification	定义:设计验证
[30]	21CFR820.30(f)	Design Controls: Design Verification	设计控制:设计验证
[31]	21CFR820.3her + 21CFR820.30(j)	Design Controls: Design History File (DHF)	设计控制:设计历史记录文件
[32]	21CFR820.3(i) + 21CFR820.184	Records: Device History Record (DHR)	记录:产品追溯性记录
[33]	21CFR820.181	Records: Device Master Record (DMR)	记录:产品制造性文档
[34]	21CFR820.30(b)	Design Controls: Design and development Planning	设计控制:设计开发计划
[35]	IEC 62304: 2006+AMD1:2015	Medical devices software-software lifecycle processes	医疗器械软件-软件生命周期开发过程
[36]	ISO 13485:2016	Medical Devices—Quality Management Systems—requirements for regulatory purposes	医疗器械-质量管理体系-用于法规的要求
[37]	21CFR820.60	Identification and Traceability: Identification	标识和可追溯性: 标识
[38]	11106101-AND-02S	QR0 - Basic Quality Requirements for Business Areas/Lines and Technology Excellence Units	QR 0 –业务领域/生产线和技术卓越单元的基本质量要求
[39]	5521922-AND-320	Software Development WI	软件开发作业指导书
[40]	5521922-AND-302	Product Place on Market in China WI	中国产品上市作业指导书
[41]	5521922-AND-319	Artificial Intelligence Medical Device Development Process	人工智能医疗器械开发流程

[42]	5521922-AND-315	Product Security Working Instruction	网络安全作业指导书
[43]	10890606-AND-E12	Data Management	数据管理
[44]	5521906-AND-215	SSME Data Privacy Handling Procedure	SSME 数据隐私处理程序
[45]	5521922-AND-254	SSME AI Data Management Work Instruction Procedure	SSME 人工智能数据管理 流程工作指导

3 Abbreviations and Definitions/ 缩略语与定义

3.1 Abbreviations/ 缩略语

AI	Artificial Intelligence	人工智能
CE	European Conformity (Conformité Européenne)	欧盟符合性认证
CFR	Code of Federal Regulations	美国联邦法规汇编
CT	Computed Tomography	计算机断层扫描
CS	Customer Service	客户服务
CUT	Clinical Use Test	临床使用测试
DHF	Design History File	设计历史记录文件
DHR	Device History Record	产品追溯性记录
DMR	Device Master Record	产品制造性文档
EM	Engineering Sample/Prototype	工程样机
EMC	Electro Magnetic Compatibility	电磁兼容性
EOD	End Of Delivery (of spare parts)	(备件)发货结束
EOS	End Of Product Support	产品支持结束
EOL	End Of Product Life cycle	产品生命周期结束
EPI	Early Procurement Involvement	早期采购参与
ERS	(system) Engineering Requirement Specification	工程需求规范
eSW	Embedded Software	嵌入式软件
ESI	Early Supplier Involvement	早期供应商参与
FM	Production Sample/Prototype	生产样机
FDA	Food & Drug Administration	美国食品和药物管理局
FMEA	Failure Modes and Effects Analysis	失效模式和效果分析
FPGA	Field Programmable Gate Array	现场可编程门阵列
FPP	Financial product plan	产品财务计划
FSP	Functional Specification	功能规范
FW	Firmware	固件
GS1	Global Standard One	(FDA 授权的 GTIN 机构)
GTIN	Global Trade Item Number	全球贸易项目代码
iCUT	Internal Clinical Use Test	内部临床使用测试
IFU	Instruction for Use	使用说明
IVK	Installed Volume Component	须跟踪的现场安装部件

Restricted

MDD	Medical Device Directive	医疗器材指令
MDR	Medical Device Regulation	医疗器械法规
MDF	Medical Device File	医疗器械文档
MQR	Medical Quality Regulation	医疗器械质量法规
MRS	Marketing Requirement Specification, Product requirement specification	市场需求规范
Mxxx	Decision Milestone M120, M150, M200, M300, M400, M500 Review, M450 (EOS), M500(EOL)	决策里程碑 M120, M150, M200, M300, M400, M450, M500 主评审
nP SW	Non-product Software	非产品软件
OTS(SW)	Off The Shelf (Software)	现成软件
OSS	Open-Source Software	开源软件
PLM	Product Lifecycle Management	产品生命周期管理
PMA	Project Milestone Approval	项目里程碑批准
PMP	Project Management Plan	项目管理计划
PSG	Product Steering Group	项目领导小组
QMP	Quality Management Plan	质量管理计划
QMSR	Quality Management System Regulation	质量管理体系法规
QR	Quality Regulation	质量保证程序
QT	Quality Technology	质量部
R1-R5	Design Reviews - R1: Product study completed, ready to start next phase. - R2: Product concept completed, ready to start next phase. - R3: Design specifications and implementation completed, ready to start verification. - R4: System successfully verified, ready to start design validation. R5: Product successfully validated, no open issues in documents, ready for product release.	设计评审 - R1: 产品研究完成, 准备开始 下一阶段。 - R2: 产品概念完成, 准备开始 下一阶段。 - R3: 设计规范和实现已完成, 准备开始验证。 - R4: 系统已成功验证, 可以启动 设计确认。 R5: 产品已成功确认, 文档中无未解 决的问题, 准备进行发布产品。
pR XX	Partial review only if hardware / embedded software is developed. The following partial reviews exist: pR DF = partial review design freeze pR IR = partial review integration release pR SF = partial review system freeze	仅当硬件/嵌入式软件开发时进行部 分审查。以下是存在的部分评审: pR DF = 部分评审设计冻结 pR IR = 部分评审集成发布 pR SF = 部分评审系统冻结
R&D	Research and Development	研发
SCM	Supply Chain Management	供应链管理
SCMP	Software Configuration Management Plan	软件配置管理计划
SSME	Siemens Shanghai Medical Equipment Ltd.	上海西门子医疗器械有限公司
SW	Software	软件
UML	Unified Modeling Language	准建模语言

WPP/GWB	Economic Product Plan/ Economic Value-Added	经营产品计划/经营增值
WI	Working Instruction	作业指导书
ZM	Drawing Sample/Prototype	图纸样机

3.2 Definitions / 定义

3.2.1 Engineering sample/prototype / 工程样机 (EM)

Also known as a test sample. Produced based on preliminary engineering documents.	也被称作测试样机，是根据初步的工程文件生产而成。
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3.2.2 Drawing sample/prototype (ZM) / 图纸样机 (ZM)

Functional sample created based on approved development documents. These documents, even in amended, incremented versions, are the basis for creating the production documents. The drawing sample is used to test and confirm the functionality of a system/component.	根据批准的开发文件生产的功能样机，即图纸样机。这些文件，即使是修订或更新的版本，也是创建生产文件的基础。图纸样机用于测试和确认系统/组件的功能。
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3.2.3 Production sample/prototype/ 生产样机 (FM)

Functional sample produced based on approved production documents and the first batches of bought-in parts from suppliers. A production review is carried out to confirm that a system/component can be reproduced. Production samples can also be produced from reworked drawing samples.	根据批准的生产文件和供应商的第一批购买零件生产的功能样机，即生产样机。进行生产审查以确认系统/组件可以复制。 如能证实，生产样机也可由图纸样机经返工而成。
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3.2.4 Formative evaluation/ 形成性评估

Process to explore user interface design strengths, weaknesses, and unanticipated use errors.	探索用户界面设计优势，劣势和意外使用错误的流程。
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3.2.5 Summative evaluation/ 总结性评估

Process to obtain objective evidence that the user interface can be used safely.	获得客观证据，以证明该用户界面可以被安全地使用的流程。
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3.2.6 eSW/嵌入式软件

a) Is deemed to be Software if CPU power is required to execute code and data using parameters and/or on a configurable basis. Data may be modified during use. Code and data can be stored on the target hardware via a data link before being used or executed (e.g. update). It is called "embedded software". b) Is deemed to be Hardware if tools (e.g., VHDL (very high-speed integrated circuit hardware description language)) are used to embed logic in the hardware in a non-modifiable and non-configurable form (i.e. code and data are not transferred to target hardware via a data link before they are used and executed) and no CPU power is required to execute this logic. It is called "FPGA".	a) 如果需要 CPU 在一定参数下，并且/或在可配置的基础上执行代码和数据，在使用过程中可能会修改数据。在使用或执行（例如更新）之前，代码和数据可以经由数据链路存储在目标硬件上。则被视为软件。称为“嵌入式软件”。 b) 如果需要使用工具（例如 VHDL（超快速集成线路硬件描述语言））以不可修改和不可配置的形式（即代码和数据不传输到目标硬件）在硬件中嵌入逻辑，在使用和执行之前通过数据链路），并且不需要 CPU 来执行此逻辑。则被视为硬件。称为“FPGA(现场可编程门阵列)”。
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3.2.7 Intended Use/ 预期用途

The intended use describes the medical purpose and/or intended medical indications, patient and intended patient population, clinical application, intended user profile and limitations of use (contra-indications and precautions).	预期用途描述医疗目的和/或预期医学适应症、患者和预期患者群体、临床应用、预期用户概况和使用限制（禁忌症和注意事项）。
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3.2.8 Intended Purpose/ 预期目的

Intended purpose means the use for which a device is intended according to the data supplied by the manufacturer on the label, in the instructions for use or in promotional or sales materials or statements and as specified by the manufacturer in the clinical evaluation.	预期目的是指设备预期用途根据制造商在标签、使用说明或促销或销售材料或声明中提供的数据以及制造商在临床评估中指定的数据。
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3.2.9 Software Component/ 软件组件

Software element that implements a defined functionality and is normally described by a design description, separation into several software units is possible for testing purposes.	实现已定义功能且通常由设计说明描述的软件元素，可以分成多个软件单元以进行测试。
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3.2.10 Non-Product Software/ 非产品软件

Computer software used in the quality management system, including software used in development, production and service provision and software used for the monitoring and measurement of requirements. nP-SW software is not part of a medical device or a medical device itself.	用于质量管理体系中的计算机软件，包括在开发、生产和服务提供中使用的软件，以及用于监控和测量要求的软件。非产品软件（nP-SW）不是医疗器械的一部分，也不是医疗器械本身。
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3.2.11 Component/组件

A component is a part of a subsystem that realizes a relevant self-contained part of a function and interacts with other components via defined interfaces. Components are integrated into subsystems.	组件是子系统的一个部分，实现了功能的有关自包含部分，并通过定义的接口与其它组件进行交互。组件被集成到子系统中。
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3.2.12 Hardware Component/硬件组件

A hardware component is an individual assembly, which is integrated with other hardware components to a hardware function block (a hardware subsystem).	硬件组件是一个单独的装配体，它与其它硬件组件一起被集成到一个硬件功能块（一个硬件子系统）中。
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4 Flow chart / 流程图

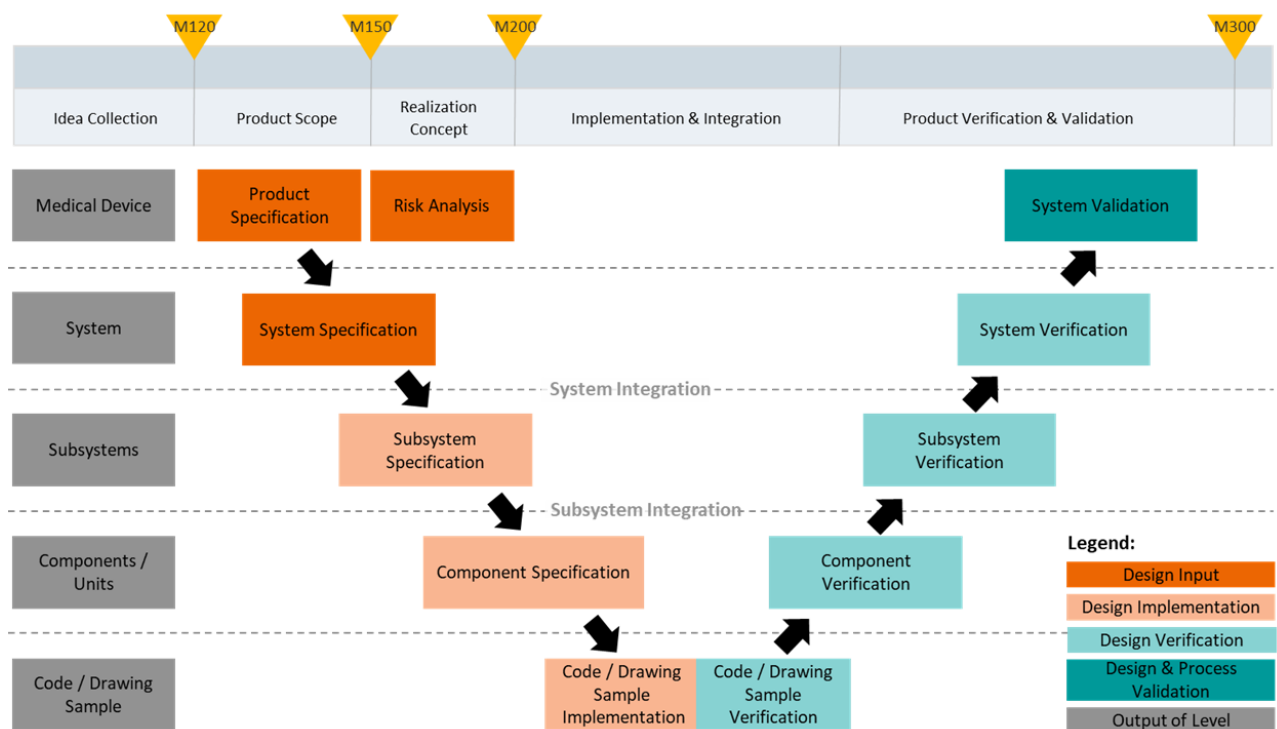
n.a.

无

5 Process & Requirement/ 过程和需求

5.1 V-model/Requirement Tracing/ V 型模式/需求跟踪

This CT QR describes the product engineering process based on the V-model of product engineering. (see Figure 1)	本质量保证程序根据产品设计 V 字型模式说明产品设计过程。见 Figure 1.
The description of product engineering process in the following chapters shows the complete procedure from the systems perspective. Sub-systems / system components similarly follow the V-model (refer to chapter 5.2).	本文以下章节所阐述的产品设计的完整步骤是针对系统的。对于子系统及系统组件，同样也要遵循这一 V 字形模式。(参照5.2节)



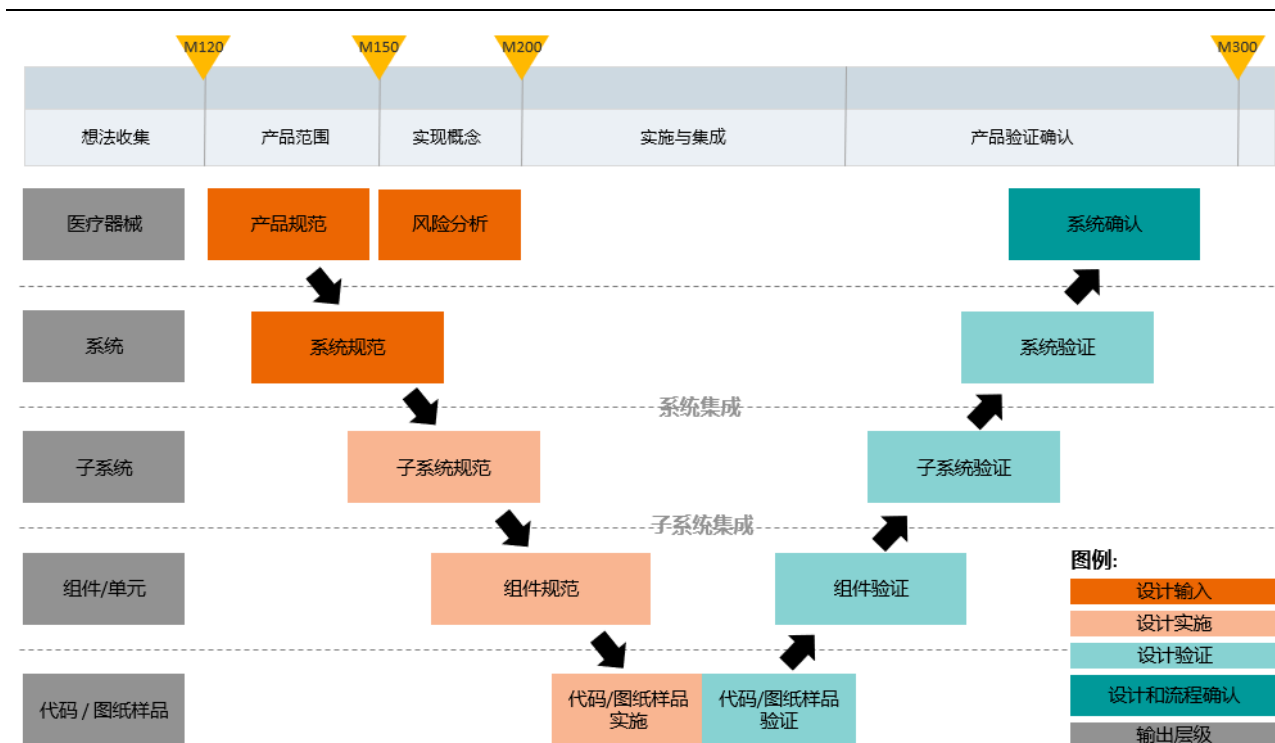


Figure 1/图 1: V-model/V 字形模式

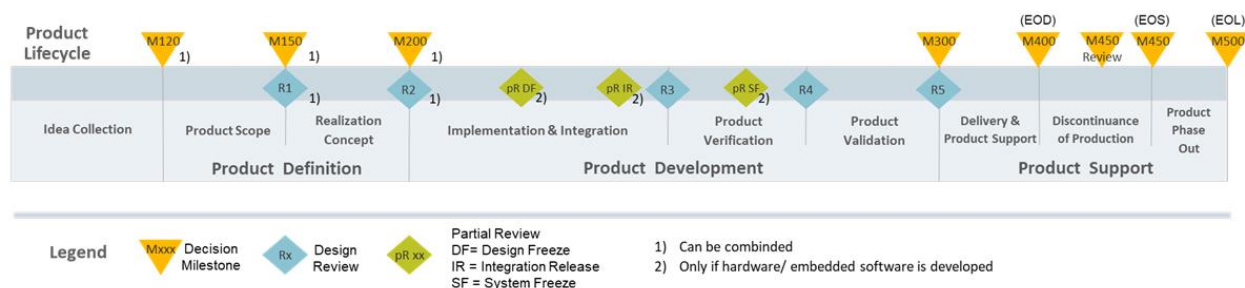
Left branch of the V-model illustrates how requirements are refined from product specification level, including the corresponding risk analysis, to system specification level, subsystem specification level through to the component specification / design description level and are implemented in a corresponding design/ code. The risk management process [8], usability engineering process [19], product security process and the S&L process (e.g. EMC, labeling) are also implemented alongside the development through all the levels, as well as the creation of the instructions for use and service documentation. Requirements are described in a structured manner. They shall be complete, unambiguous, able to be verified or validated, and not in conflict with each other. They should contain acceptance criteria. Specifications for implementation should only be described as detailed as necessary, to give the responsible competence centers or suppliers enough flexibility. The requirements from all requirement levels up to the testing levels must be demonstrably traceable through the development documents, vertically and horizontally. Trace results are documented. To trace the requirements, the relevant documents use structuring, grouping or special identifiers to illustrate how the requirements are converted into a solution. The technical review ensures that the requirements are sufficiently and clearly formulated and that the requirements have been fully and accurately translated into verifiable specifications.	V 字型模式左侧说明了需求是如何从产品需求规范层面细化到系统需求规范层面，再到子系统需求规范层面，直至组件需求规范/设计描述层面，并通过相应的设计/代码实现的，包括对应的风险分析。 风险管理流程[8], 可用性工程流程[19], 产品安全流程以及法规流程（例如 EMC、标签）也伴随着开发在所有层面上实施，同时创建使用说明书和服务文档。 需求应以结构化的方式描述。它们必须是完整的、无歧义的、可验证或可确认的，并且彼此之间不冲突。它们应包含验收标准。实施规范的描述应仅详细到必要的程度，以赋予负责的能力中心或供应商足够的灵活性。 从所有需求层级的需求到测试层级的需求必须能够通过开发文档在垂直和水平方向上得到可证明的追溯性。 追溯结果被记录在案。 为了追溯需求，相关文档使用结构化、分组或特殊标识符来说明需求是如何转化为解决方案的。 技术评审确保需求被充分且清晰地制定，并且需求已被完整且准确地转化为可验证的规范。
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The right branch of the V-model provides on each test level sufficient and adequate test cases to verify/validate the requirements on the left side of the V-model. The documentation to be created can be specifically tailored, depending on the technology (mechanical, electromechanical, electronic, software). Components are developed/ implemented and integrated to subsystems. After subsystem integration the verification of the subsystem is performed. Subsystems are integrated to a system. After system integration the system verification is performed. The relevant test levels are opposed against the requirements based on a test concept. Predefined test specifications are used in the corresponding test levels to demonstrate that the requirements from the different levels have been met. The detailed assignment of tests to the requirements is shown in the test plans/test specifications. For the product's safe and proper use all design outputs are essential.	V 字型模式右侧 在 V 模型的每个测试层级提供足够和适当的测试用例,以验证/确认 V 模型左侧的需求。 需要创建的文档可以根据不同技术（机械、机电、电子、软件）特别定制。 组件被开发/实现并集成到子系统中。子系统集成后,执行子系统的验证。 子系统被集成到系统中。系统集成后,进行系统验证。 相关的测试层级根据测试概念与需求相对应。 在相应的测试层级中使用预定义的测试规范来证明不同层级的需求已得到满足。测试到需求的详细分配显示在测试计划/测试规范中。为了产品安全和正确使用,所有的设计输出都是必不可少的。
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Design Input Design input means the physical and performance requirements of a device that are used as a basis for device design. The design input requirements relating to a device are appropriate and describe the intended use/intended purpose of the device, including the needs of the user and patients. They shall include: a. functional, performance, usability, and safety requirements, according to the intended use/intended purpose; minimum: use scenarios, market requirement specification, engineering requirement specification b. applicable regulatory requirements and standards; minimum: country requirements in market requirement specification and standards in engineering requirement specification c. applicable output(s) of risk management; minimum: risk analysis d. as appropriate, information derived from previous similar designs; see project quality and management plan for implementation e. other requirements essential for design and development of the product and processes (e.g. Manufacturability, Serviceability). Design input requirements are documented and reviewed for completeness, unambiguousness, and consistency (non-conflicting). They are approved by a designated individual. The approval, including the date and signature of the individual(s) approving the requirements, is documented. For further details to Design Input see [22], [23], [36].	设计输入 设计输入是指作为设备设计基础的设备物理和性能要求。 与器械相关的设计输入需求是适当的，并描述了器械的预期用途/目的，包括用户和患者的需求。它们应包括： a. 功能、性能、可用性和安全要求，根据预期用途/预期目的；最小：使用场景、市场需求规范、工程需求规范 b. 适用的监管要求和标准：最低：市场需求规范中的国家要求和工程需求规范中的标准 c. 风险管理的适用输出：最低：风险分析 d. 酌情，从以前的类似设计中获取信息；查看项目质量和实施管理计划 e. 产品和工艺的设计和开发所必需的其他需求（例如：可制造性，可服务性） 设计输入要求记录并审查完整性、明确性和一致性（无冲突）。它们由指定的个人批准。批准，包括批准需求的个人的签名和日期，被记录在案。 有关设计输入的更多详细信息见[22]，[23]，[36]。
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Design Output Design output means the results of a design effort at each design phase and at the end of the total design effort. The finished design output consists of the device, its packaging and labeling, and the device master record. The development output is defined and documented in a form that facilitates an adequate evaluation of conformance to design input requirements. Design and development outputs shall: - meet the input requirements for design and development; minimum: test summary reports - provide appropriate information for purchasing, production and service provision; minimum: design transfers to suppliers (e.g. purchasing information), production and service - contain or reference product acceptance criteria; minimum: design review records - specify the characteristics of the product that are essential for its safe and proper use. minimum: general safety and performance requirements (according to Medical Device Regulation, former Medical Device Directive): essential requirement document The outputs of design and development shall be in a form suitable for verification against the design and development inputs and shall be approved prior to release to customer. Records of the design and development outputs shall be maintained. Design outputs ensure the proper functioning of the device. They are documented, reviewed, and approved before release. The approval, including date and signature of the individual(s) approving the output, is documented. For further details to Design Input see [24], [25], [36].	设计输出 设计输出指的是每个设计阶段的设计工作结果，以及整个设计工作结束时的结果。最终的设计输出包括设备本身、其包装和标签，以及设备主记录。 开发输出的定义和记录形式有助于对符合设计输入要求的合规性进行充分评估。 设计和开发输出应： - 满足设计和开发的输入需求：最低：测试摘要报告 - 提供适当的采购、生产和服务信息：最低：向供应商转让设计（如采购信息）、生产和服务 - 包含或参考产品验收标准：最低：设计审查记录 - 指定产品对安全和适当使用至关重要的特性。最低：一般安全和性能要求（根据医疗器械条例，前医疗器械指令）：基本要求文件 设计和开发成果应当以适合对设计开发输入进行验证的形式，并在发布给客户前批准。 应保留设计和开发输出的记录。 设计输出可确保设备正常运行。它们在发布前被记录、审查和批准。批准，包括批准产出的个人的日期和签名，记录在案。 有关设计输出的更多详细信息见 [24], [25], [36].
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5.2 Structure of the Product Engineering Process/ 产品设计过程结构



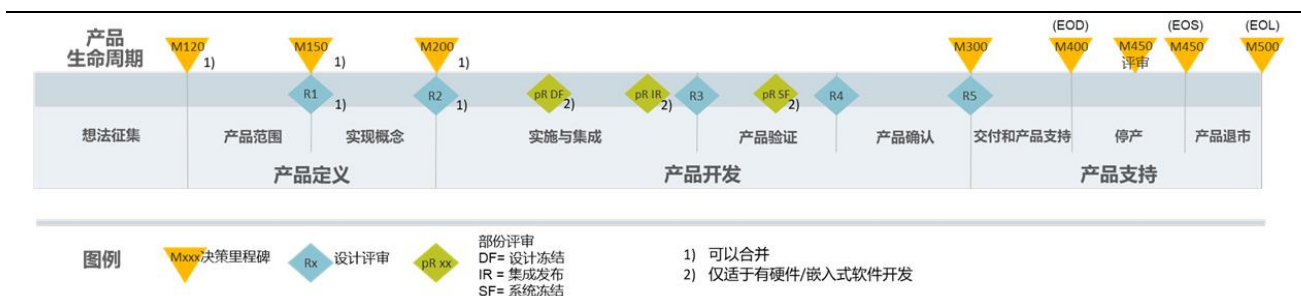


Figure 2/图 2: Structure of the product engineering process/ 产品设计过程结构

The development of products is carried out in projects, starting with the first section “Idea collection”. At integration points components are integrated into subsystem and subsystems are integrated into the system. Software subsystems are defined and created according to [39], embedded and approved in the system project (delivery release).	产品开发在项目中进行, 从第一阶段“想法征集”开始。 在集成点, 组件被集成到子系统和子系统被集成到系统中。软件子系统根据[39] 定义和创建, 被嵌入并批准在系统项目(交付发布)中。
The design reviews pR DF, pR IR, pR SF only occur if hardware / eSW are developed. On demand the sequence of the phases and milestones is stated precisely within the quality management plan.	设计评审 pR DF, pR IR, pR SF 只在有硬件和嵌入式软件开发的项目中需要, 并在相关项目的质量管理计划中准确定义。
To ensure that the product engineering process is transparent and manageable in terms of the process goals for quality, time, and costs, it is split into different phases, as shown in Figure 2. Each phase comprises of: - Decision to start - Work out results - Design review(s) on completion	为确保产品工程过程透明, 并按照质量、时间和成本的过程目标进行管理, 将产品过程分割为不同阶段, 如图2所示。每个阶段包括: - 启动决定 - 设计实现 - 完成工作的设计审查
Design review R4 verifies that the development and production output match the requirements (verification). Design review R5 verifies that the design validation has been successfully performed.	设计评审 R4 验证开发和生产输出与需求相匹配 (验证)。设计评审 R5 验证已成功执行设计确认。
For tailoring rules see section 6.2	有关定制规则, 请参见第 6.2 章节

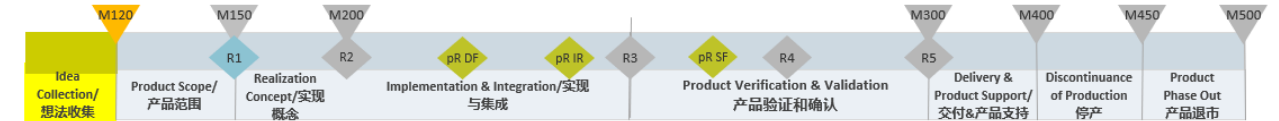
5.3 System development phases and their purpose/ 系统开发阶段及其目的

The details below describe the sections of the development process and their purpose in chronological order. The end-of-phase output required for the design reviews is described in [13] or [14]. [13] is dedicated used for SOMARIS 5 project. [14] is dedicated used for SOMARIS10 Project. In the following sub chapters, the most relevant <i>end-of-phase outputs(deliverables)are formatted italic.</i>	如下细节按照时间顺序描述了开发过程的阶段及其目的。设计审查结束时的输出需求在[13]或[14]中描述。 [13] 仅用于 SOMARIS 5 的项目。 [14] 仅用于 SOMARIS 10 的项目。 在以下子章节中关于 <i>阶段末的输出 (交付件)</i> 标记为斜体字。
The decision to start a new phase is based on the outcome of the design review.	是否决定开始新阶段基于设计审查的结果。
Decisions are approved on project phase applications (appendix 2).	在项目阶段应用中批准决定 (附录 2)。
Design reviews may be conducted for systems, subsystems, or components.	设计审查可对系统、子系统或组件进行。

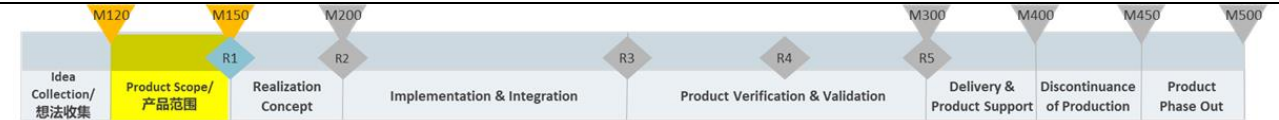
Restricted

Specific information for when the work product must be finalized, archived, and the applicable review rules can be found in the reference data and the input for milestones	工作产品必须完成、归档的具体信息，以及适用的审查规则，可以在参考数据和里程碑的输入中找到。
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5.3.1 Process Phase 1: Idea Collection / 过程阶段 1: 想法收集

	
Idea Collection (up to M120)	想法收集（截止到M120）
- Collection and evaluation of product ideas. - Analysis of product-specific market - Specification of business plan	- 产品想法收集和评估。 - 产品特定市场分析 - 商业计划书的具体规范或详细说明

5.3.2 Process Phase 2: Product Definition / 过程阶段 2: 产品定义

	
Product scope (M120 – M150)	产品范围（M120– M150）
Start: M120 Decision	开始: M120 决定
Project request M120 incl. assessment of economic feasibility.	项目要求M120包括对经济可行性的评估。
Purpose of the process phase	流程阶段的目的
- Definition of the product - Determination of market, customer, service, and SCM/ manufacturing requirements - Carry out product strategy and product profile planning - Where applicable, front-end trials, technology developments, clinical studies	- 产品定义 - 确定市场、客户、服务以及供应链管理SCM/ 制造要求 - 执行产品策略和产品画像（综合特征描述）规划 - 在适用的情况下，进行前端试验、技术开发和临床研究
Activities and core artifacts (that are started during this phase)	活动和核心产出物 (在此阶段开始的)
- Defining the Product profile by determining, defining, and prioritizing: - Market, customer, service, and SCM/manufacturing requirements, together with their systematic translation into hardware features (system functions) and software features (software functions). - Historical data for predecessor products, - Technical dependencies between hardware, eSW and software including all available platforms are evaluated.	产品画像: 通过确定，定义和优先排序来定义。 - 对市场、客户、服务和供应链管理/ 制造需求，以及将其系统的转化为硬件指标（系统功能），软件指标（软件功能）。 - 此前产品的历史数据， - 对硬件，嵌入式软件和系统软件之间的技术依赖关系，包括所有可用平台的技术相关性进行评估。
Defining the Product concept: by applying methods that take market and customer requirements into consideration for reaching the goals of:	产品概念: 通过采用考虑市场和客户需求的方法来定义，以达到： - 实现战略业务目标， - 缩短计划时间，更好的质量规划，

○ implementing strategic business objectives, ○ shorter planning times with a better planning quality, ○ a more efficient definition of product and system requirements. ● Defining the Product strategy (recommended) by: ○ Do a market survey to analyze, amongst others, the potential and structure of the market (market volume, homogeneity, accessibility) and target groups (segmentation), ○ By analyzing and defining the key success factors of the new product, the potential and structure of the market (market volume, homogeneity, accessibility), the target groups (segmentation), its strengths, and weaknesses as well as restrictions of the organization. ○ Take into consideration existing strategic technology roadmaps as well as predefined strategies to create synergies.	○ 更有效的产品和系统需求定义。 ● 产品策略: (推荐通过以下定义) ○ 做市场调查, 分析市场潜力和结构 (市场容量, 均匀性, 可及性) 和目标群体 (市场划分)。 ○ 分析和定义新产品的关键成功因素, 市场潜力和结构 (市场容量、同质性、可接近性), 以及目标群体 (市场细分), 优势和劣势, 以及该组织的限制。 ○ 统筹考虑现有的战略技术路线图, 以及预定义的策略, 以创造协同效应。
● If necessary, feasibility tests, studies, and technology developments are carried out, and industrial design (only if hardware system is developed) is clarified. Results from these activities may be described in a system implementation concept. Results relating to industrial design must be summarized.	● 如有必要, 进行可行性试验、研究和技术开发, 对工业设计 (仅当开发硬件系统时) 进行澄清。这些活动的结果可以记录在系统实现概念文档中。必须对工业设计相关的结果进行总结。
● Economic feasibility assessment is conducted to evaluate the project's financial viability.	● 进行经济可行性评估以评估项目的财务可行性。
● Specify product scope: <i>market requirement specification (MRS)</i>, <i>use scenarios</i>, and <i>use specification</i>.	● 明确产品范围: <i>产品市场需求规范 (MRS)</i>, <i>使用场景</i> 和 <i>使用规范</i>。
● Project organization and responsibilities are specified in the <i>project management plan (PMP)</i>; persons are assigned to roles (as needed for the next project phase).	● 在 <i>项目管理计划</i> 中明确项目组织和职责, 分配人员角色 (为下一项目阶段做准备)。
● A project-specific <i>quality management plan (QMP)</i> is created and maintained for each project. Approval / termination criteria for individual project phases are defined.	● 创建并维护针对每个项目的 <i>质量管理计划</i>。定义各个项目阶段批准/终止标准。
● Create the <i>usability engineering file</i> (see section 7.18)	● 创建 <i>可用性工程文件</i> (见 7.18)
● Preventive measures to avoid faults/errors are identified and documented in the <i>error prevention report</i> (see section 6.11).	● 确认预防性措施, 避免故障/错误 并且在 <i>错误预防报告</i> 中记录 (见 6.11)。
● A product specific <i>reliability engineering plan</i> is created.	● 创建产品特定 <i>可靠性工程计划</i>。
● Compatibility of customer requirements with standards and statutory provisions is checked.	● 检查客户需求与标准和法规条款间的兼容性。
● Assess if AI is in scope and plan for	● 评估 AI 是否属于适用范畴, 如适用, 则制

compliance with AI requirements if applicable - It is assessed whether any AI systems are included in the product or its components. If so, planning begins for compliance with AI regulations, including identifying high-risk systems and considering further regulatory controls for later phases. - If the product includes innovative elements (e.g. new materials, design, or intended use), assess whether clinical data is required to demonstrate safety and performance. This applies regardless of risk class. For high-risk products with unique design-performance dependencies, clinical data may also be required. The need for clinical investigation must be evaluated early and documented accordingly.	定满足 AI 要求的合规计划 - 评估产品或其任何组件中是否包含人工智能系统；若包含，则立即启动人工智能法规合规规划，包括识别高风险系统，并为后续阶段考虑进一步的监管控制措施。 - 若产品包含创新性要素（例如新材料、新设计或新预期用途），则须评估是否需要临床数据来证明其安全性与性能，这一要求适用于所有风险等级。对于因独特设计而存在特殊性能依赖的高风险产品，也可能需要临床数据。必须尽早评估是否需开展临床研究，并予以相应记录。
- Assess relevance of EU Data Act for product data - The fundamental right to the protection of personal data is addressed through [44]	- 评估《欧盟数据法案》（EU Data Act）对产品数据的适用性 - 个人数据保护的基本权利已得到阐述[44]
Realization concept (M150 – M200)	实现概念（M150– M200）
Start: M150 Decision	开始: M150 决定
- Design Review R1 - Project request M150 incl. FPP	- 设计评审R1 - 项目要求M150包括FPP
Purpose of the process phase	流程阶段的目的
- Translation of product requirements into technical system requirements. - Definition of the system architecture - Identification of the product's potential hazards	- 将产品需求转化为技术系统需求 - 定义系统架构 - 识别产品潜在的危害
Activities and core artifacts (that are started during this phase)	活动和核心产出物 (在此阶段开始的)
Specify system realization concept	明确系统实现概念
Market, customer, and service requirements are translated into system/platform development requirements: <i>engineering requirement specification (PRS/ERS)</i>.	将市场、客户和服务需求转化为整个系统/平台开发需求工程需求规范（PRS/ERS）。
User interface specification is created.	创建用户界面规范。
Specify High-level system and software architecture are defined (e.g. based on increments produced as prototypes). Components and their interfaces, together with the external interfaces to other components, are defined.	定义高级别系统和软件架构（例如，基于增量的制作原型）。定义组件及其接口，以及与其他组件间的外部接口。
Specify subsystem architecture, requirements, and realization concept: <i>software architecture</i>.	指定子系统架构、需求和实现概念：软件架构。
Software safety classes are assigned to the	根据架构及规定的相关活动（包括隔离，

components identified by the architecture and any necessary associated action specified (including segregation, if appropriate). (see [39])	如果合适的话)分配部件的软件安全等级。见[39]
Activities to ensure that software elements can be reproduced are defined, set up and described in <i>software configuration management plan</i>.	确保软件元素可以复制的活动在<i>软件配置管理计划</i>中定义、设置和描述。
Product risks are identified and assessed and documented in the <i>risk analysis</i>; action plans to reduce the risk are defined (see [8]).	识别和评估产品风险；定义降低风险的行动计划（见[8]）并在<i>风险分析</i>中记录。
Identify and evaluate potential security risks for the product: <i>threat and risk analysis</i>.	识别和评估产品潜在的安全风险：<i>威胁与风险分析</i>。
Determine security requirements: <i>security requirement specification</i>	确定安全要求：<i>安全需求规范</i>。
Determine device classification for target markets (e.g. EU MDR and EU AI Act)	确定目标市场的设备分类（EU MDR 和 EU AI Act）。
Plan tests for verification and validation: <i>test concept</i>	规划验证确认测试：<i>测试概念</i>
Plan clinical development and - evaluation: <i>clinical development plan, clinical evaluation plan</i>	规划临床开发和评估：<i>临床开发计划，临床评估计划</i>。
Assess claims and plan evidence: (<i>initial claim list</i>)	评估声明并规划证据：<i>（初始）声明清单</i>。
Technical feasibility/prototyping (e.g. Engineering samples) and the possible reuse of existing components is assessed.	对技术可行性/原形（如工程样品）和对现有组件的可能重用进行评估。
Decisions are taken on whether the development is going to be in-house and/or carried out by third parties; this includes selection of suppliers (see [2]).	决定是否由内部和/或第三方执行开发；包括选择供应商（见[2]）。
- Strategic procurement must be involved as early as possible to follow-up on the supplier technologies at the best, and to optimize the cost position for the development process. - The Early Procurement Involvement (EPI) and Early Supplier Involvement (ESI) is required, or when design decisions must be agreed with suppliers.	- 战略采购必须尽早介入，以便及时跟进供应商技术，并优化整个开发过程的成本结构。 - 需要早期采购参与（EPI）和早期供应商参与（ESI），或者当设计决策必须与供应商达成一致时。
Planning is carried out for country-specific approvals; requirements arising from standards and statutory provisions (S&L) are identified and reviewed.	针对具体国家审批做出计划；识别和审查标准和法律法规（S&L）提出的要求。
The project planning is finalized, the quality and project management plans are updated, if necessary. The compliance with the approval criteria in the quality management plan is reviewed at every design review.	完成项目计划，更新质量和项目管理计划。在每次设计评审时审查质量管理计划中对批准标准的符合性。
Identify prohibited AI practices, classify AI system risk and plan for data governance. See [41].	识别被禁止的人工智能行为，对人工智能系统进行风险分级，并规划数据治理方案。见[41]。
Plan for compliance with EU Data Act (data	制定《欧盟数据法案》（EU Data Act）合

availability, sharing, design)	规划计划（涵盖数据可用性、数据共享与产品设计）
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5.3.3 Process Phase 3: Product Development / 过程阶段 3：产品开发

Implementation & Integration (M200 – R3)	实现&集成 (M200– R3)
Start: M200 Decision	开始: M200 决定
- Design Review R2 - Project request M200 incl. FPP	- 设计评审R2 - 项目要求M200包括FPP
Purpose of the process phase	流程阶段的目的
- Development of the product (definition of design, implementation, verification) - Translation of requirements from subsystem requirement specification(s) into component requirement specifications, design descriptions, and designs	- 产品开发（设计定义、实现、验证） - 从子系统需求规范中将需求转化为组件需求规范、设计描述和设计
Activities and core artifacts (that are started during this phase)	活动和核心产出物 (在此阶段开始的)
The requirements from the subsystem requirement specification(s) are translated into <i>components requirements specifications, design specifications, and designs</i>.	将子系统需求规范转换成 <i>部件需求规范, 设计规范</i> 和 <i>设计实现</i>。
The requirements for the contractor must then be specified, depending on the decision as to whether the development is going to be in-house and/or carried out by suppliers.	基于由内部和/或供应商执行开发的决定, 相应的需求必须描述清楚。
Country-specific instruction for use is prepared, and the requirement specification is created.	准备针对特定国家的使用说明并创建需求规范。
Service support documentation is prepared.	准备服务支持文档。
Continuation of the risk management activities	继续进行风险管理活动
Specify verification activities: <i>system verification test plan</i>	明确验证活动: <i>系统验证测试计划</i>
Create detailed product design for HW and SW	为硬件 (HW) 和软件 (SW) 创建详细的产品设计
Implement and realize HW and SW design	实施和实现硬件 (HW) 和软件 (SW) 设计
Create test specifications, perform verification test at unit/component levels and document verification results; create the necessary detailed design description: <i>test specification, test record, design description</i>	创建测试规范, 执行单元/组件级别的验证测试并记录验证结果; 创建必要的详细设计说明: <i>测试规范、测试记录、设计描述</i>。
Ensure data quality and documentation for AI and prepare user information. See [43], [45].	确保人工智能所用数据的质量与文档记录, 并准备面向用户的信息。见 [43], [45]。
Additionally for Hardware Implementation (M200 – pR DF)	硬件实现补充 (M200 – pR DF):

• A serial number range is specified.	• 指定一个序列号范围。
• Description of possible system configurations (combinations of hardware and software configurations) is drafted.	• 提出可能的系统配置（结合硬件和软件配置）描述的初稿。
• Spare parts are specified.	• 对备件进行说明。
• <i>Mechanical design</i> is concluded (design freeze) and the <i>FPGA design</i> is specified as the prerequisite for carrying out component tests.	• 作为进行组件测试的前提条件，总结 <i>机械设计</i>（设计定型），说明FPGA（<i>现场可编程门阵列</i>）设计。
Additionally for Software Implementation (M200 – pR IR/R3)	软件实现补充 (M200 – pR IR/R3)
• Software functionality is implemented on an iterative basis guided by the requirements in the software subsystem requirements specification up to the point at which the software is ready to be integrated into the system. This includes integrating the components, carrying out a successful verification process at unit and component levels, creating the necessary detailed design documentation (detailed functionality, detailed architecture, testing specifications / automated testing), and preparing the user documentation.	• 软件功能是在软件子系统需求规范引导的迭代算法的基础上实现的, 直到该软件已准备好被集成到系统中。这包括集成组件, 在单元和组件层面成功执行验证过程, 创建所需的详细设计文档（详细功能和架构, 测试规范/自动化测试），并准备用户文档。
• From M200 onward, a configuration management system must be used for software and embedded software.	• 从M200开始，必须对软件和嵌入式软件使用配置管理系统。
• Perform SW code reviews: <i>code review summary report</i>	• 执行软件代码审查： <i>代码审查总结报告</i>
• Software is released by <i>software release note</i>.	• 软件通过 <i>软件发行说明</i> 发布。
• Result is the corresponding <i>test documentation</i>	• 最终生成相应的 <i>测试文档</i>
At pR IR the software and embedded software is ready for system integration (only if hardware system is developed).	在pR IR软件和嵌入式软件可以进行系统集成（仅当有硬件系统开发时）。
Integration (pR DF – pR IR) - (only if hardware system is developed)	集成 (pR DF-pR IR)（仅当有硬件系统开发时）
• Hardware components are verified and integrated into subsystems.	• 验证硬件部件并将其集成到子系统中。
• Hardware components and subsystems are accepted (partially acceptance possible at pR IR, complete acceptance at pR SF latest).	• 硬件部件和子系统可接受（在 pR IR可部分接受，最晚于pR SF完整接受）
• Result is the corresponding <i>test documentation</i>	• 最终生成相应的 <i>测试文档</i>
• The <i>system flow chart</i> is written	• 编写 <i>系统流程图</i>
• Create system stability test plan	• 创建系统稳定性测试计划
The individual subsystems have reached the level of quality required.	各个子系统满足自身的质量标准。
System integration (pR IR – R3) - (only if hardware system is developed)	系统集成 (pR IR-R3)（仅当有硬件系统开发时）
• Subsystems are integrated into the system.	• 将子系统集合到系统中。

Product verification (R3– R4)	产品验证（R3–R4）
Start: After Design Review R3	开始： 设计评审R3之后
Design Review R3	设计评审R3
Purpose of the process phase	流程阶段的目的
Proof of realization of technical requirements. The system and its documentation including accompanying documents are ready for product verification.	技术要求实现的证明。系统及其文档，包括随附文件已准备好进行产品验证。
Activities and core artifacts (that are started during this phase)	活动和核心产出物 (在此阶段开始的)
Description of possible system configurations (combinations of hardware and software configurations) is approved.	批准可能的系统配置（结合硬件和软件配置）描述。
Options for combining software configurations are verified, taking into consideration the necessary hardware requirements.	考虑到必要的硬件需求，需要结合软件配置选项进行验证。
The system verification (verification of system requirements and on demand also software requirements) is performed on drawing samples.	在图纸样机上进行系统验证(验证系统需求和按需验证软件需求)。
Perform final verification (system verification): <i>Verification test reports</i> (depending on test concept: reports for subsystem/system level; including reports for safety tests and reliability tests)	执行最终验证（系统验证）：<i>验证测试报告</i>（根据测试概念：子系统/系统级别的报告；包括安全测试和可靠性测试的报告）。
<i>Verification results</i> are summarized.	总结<i>验证结果</i>。
Failures are rectified.	纠正错误。
Create test specifications for validation: <i>system validation specification</i>	创建确认测试规范：<i>系统确认规范</i>
Action is taken to ensure prerequisites for subsequent proof of delivery are in place.	采取措施以确保满足后续发货的前提条件。
Prepare necessary production documentation (e.g. work plan, test instructions including documentation for production and test equipment).	准备所需生产文件（如工作计划，包括生产和测试设备在内的测试指令）。
Complete Country-specific <i>instructions for use</i> in the required languages.	完成具体国家所需语言的<i>使用说明</i>。
Testing on the production sample (first batch) as required for country-specific approvals (e.g. proof of compliance with safety requirements) is carried out.	按照具体国家准入要求（如安全要求合规性证明），对生产样本（第一批次）进行测试。
At pR SF (only if hardware system is developed):	在pR SF点：（仅当有硬件系统开发时）
Complete acceptance of hardware components. Transfer of individual components and/or subsystems to production is finished (<i>hardware transfer protocol</i>) (see section 6.13).	硬件组件的完整接受。各个组件和/或子系统的生产转移完成(<i>硬件转移报告</i>)(见 6.13)。
Continuance of system integration and	持续系统集成并编写<i>系统集成总结报告</i>。

write <i>system integration summary report</i> .	
The finished software verification ensures that software components are successfully integrated and work as specified.	软件验证完成，确保软件组件成功地整合并能按规定运行。
Product validation (R4 – M300)	产品确认 (R4–M300)
Start: After Design Review R4	开始: 设计评审R4之后
Design Review R4	设计评审R4
Purpose of the process phase	流程阶段的目的
- Proof of realization of product (/ technical) requirements. - Final evaluation of usability and the specified intended use/intended purpose of product. - The system and its documentation including accompanying documents are ready for product validation.	- 产品 (/技术) 要求实现的证明。 - 对产品可用性的最终评估以及对产品指定的预期用途/预期目的的评估。 - 系统及其文档，包括随附文件，已准备好进行产品确认。
Activities and core artifacts (that are started during this phase)	活动和核心产出物 (在此阶段开始的)
Design is validated (see section 6.19), including a final review of the effectiveness of the required measures derived from the risk analysis: <i>system validation test report, risk management report, formative and summative user interface evaluation report</i>.	对设计进行确认（见第6.19部分），包括风险分析所要求的防范措施的有效性最终评审：<i>系统确认测试报告，风险管理报告，形成性与总结性用户界面评估报告</i>。
Reliability measures defined in <i>reliability engineering plan</i> are implemented and verified. <i>Reliability engineering report</i> is written see ([38])	<i>可靠性工程计划</i>中定义的可靠性措施得到实施和验证。编写<i>可靠性工程报告</i>见 ([38])。
<i>Production documentation</i> (e.g. work plans, test instructions including documentation for production and test equipment) is completed.	完成生产文档（如工作计划，包括生产和测试设备在内的测试指令）
<i>Series production samples</i> are created based on the approved production documents (including process validation) as specified in [4].	如[4]说明，基于已批准的生产文档（包括过程确认），创建<i>批量生产样品</i>。
<i>Country-specific accompanying documents</i> (see [12]) and translation into required languages is completed.	完成具体<i>国家要求的配套文档</i>（见[12]）及所需语言的翻译。
<i>Service support documentation</i> is completed.	完成<i>服务支持文档</i>。
Spare parts are available for delivery.	配件可发货。
<i>Validation and usability evaluation results</i> are summarized.	总结<i>验证结果和可用性评估结果</i>。
System stability test summary report is written.	编写系统稳定性测试总结报告。
Transfer of software and embedded software is finished (<i>software transfer protocol</i>).	完成软件和嵌入式软件的转移（<i>软件转移报告</i>）。
Technical documentation (<i>technical file</i>) is completed as well as the <i>application service</i>	完成技术文档（<i>技术文件</i>）及<i>应用服务策略</i>，<i>DHF</i>，<i>DHR</i>，<i>MDF</i>，<i>DMR</i>，<i>目标市场临床评</i>

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strategy, DHF, DHR, MDF, DMR, clinical evaluation for target markets, essential requirement document, S&L-reports, conformance proofs, usability engineer file and risk management file.	估, 基本要求文档, 法律法规报告, 符合性证明, 可用性工程文件和风险管理文件。
Create general safety and performance requirements (GSPR) proof of conformity.	创建通用安全和性能要求 (GSPR) 的符合性证明。
Create summary technical documentation (STED)	创建技术文件概要 (STED)
Complete clinical evaluation for the release of the medical device and define post-market surveillance activities: <i>clinical evaluation report</i>	完成医疗器械上市前的临床评价并定义上市后监管活动: <i>临床评价报告</i>。
Prepare input for post-market surveillance plan and periodic safety update report	准备上市后监管计划和定期安全更新报告的输入资料。
Further details see [20].	进一步细节参见[20]。
- Country-specific delivery approval list is updated by the following units based on the available information. - Quality management: based on available approvals and releases - Customer Documentation Team: based on the specified target market – all languages of the instruction of use - SCM: all SAP data for logistic purpose is available.	- 由如下部门基于可用信息, 更新具体国家交付审批列表。 - 质量管理: 基于现有批准和发布 - 客户文档组: 基于指定的目标市场 – 所有语言的使用说明 - SCM: 生产相关的文档必须在SAP中存档。
Compliance with the approval criteria in the quality management plan is reviewed.	审查与质量管理计划中批准标准的合规性。
The outcome of the validation activities may lead to the creation of a restriction list showing requirements that have not been implemented and/or that are insufficient present in the product.	确认活动的结果可能导致限制性列表的创建, 列出产品中未实现的和/或不足的需求。
The product has received a <i>delivery release, technical documentation, and approval documents</i> (e.g. CE-declaration) are available according to the approval plan.	产品已按照批准计划获得了交付放行, 技术文件和批准文件 (如CE-声明) 已完成。

5.3.4 Process Phase 4: Product Support / 过程阶段 4: 产品支持

Delivery & Product support (M300 – M400)	交付和产品支持 (M300 – M400)
Start: M300 Decision	开始: M300 决定
- Design Review R5 - Project request M300 incl. FPP	- 设计评审R5 - 项目要求M300包括FPP
Purpose of the process phase	流程阶段的目的
- Delivery of error free products - Product support - Preparation of the phase out	- 无缺陷产品交付 - 产品支持 - 产品退市准备

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Activities and core artifacts (that are started during this phase)	活动和核心产出物 (在此阶段开始的)
The approved product is delivered.	交付经过批准的产品。
Applications for further country-specific approvals are submitted; country approval list is updated.	递交符合特定国家批准的进一步申请; 更新批准清单。
Ensuring of support, maintenance, and service.	确保支持, 维护, 和服务。
Ensure post market surveillance and risk management.	确保后市场监管和风险管理。
Changes (e.g. replacements, enhancements) must be initiated as specified in [1].	更改 (例如替换、增强) 必须按照 [1] 中的规定进行。
It is required to evaluate the impact of discontinuation on public health according to EU MDR 10 a.	根据欧盟医疗器械法规 (EU MDR) 第10a条的要求, 对公共卫生供应中断或停止的影响进行评估。
The M400 decision may relate to a product, ranges of serial numbers of a product or a specific software version of a product.	M400决定, 可能和产品, 产品序列号范围或产品特定软件版本相关。
Further details see [39].	进一步细节参见[39]。
Discontinuation of production (M400 – M450)	生产停止 (M400 – M450)
Start: M400 Decision	开始: M400 决定
Project request M400	项目要求M400
Purpose of the process phase	流程阶段的目的
- Phase out of a product is initiated - End of support is reached	- 启动产品退市 - 停止产品支持
Activities and core artifacts (that are started during this phase)	活动和核心产出物 (在此阶段开始的)
Production of the product is discontinued.	产品的生产停止
Marketing activities are discontinued, and the product is deleted from the price book.	市场活动停止, 产品从价格书上删除。
Product support, maintenance, service support, and the provision of spare parts is guaranteed up to the M450.	产品支持, 维护, 服务支持, 和备件供应一直保证到M450.
Changes (e.g. replacements) must be initiated as specified in [1].	更新 (替换) 必须按照说明[1]进行。
The preliminary date for M450 (EOD) is determined during M400 decision.	M450的初始日期 (EOD), 决定于M400决策过程中。
After M400 regular phase-out reviews must occur by the business line and customer service at least once a year, to align the installed base reduction with service part stock levels, and to refine the planning for the final M450 date. The final M450 date (EOS) is determined by CT during the M450 review.	M400后, 业务部门和客户服务至少每年须执行一次常规的退市评审, 从而确保装机量的减少与维修零备件的库存水平相匹配, 并为最后的M450日期完善计划。最终的M450日期 (EOS), 在M450评审中, 由CT决定。

If delivery is resumed: If product delivery is resumed after having been discontinued, all the delivery requirements need to be checked and, if necessary, restored and/or a R5 design review repeated. A new M300 decision on release of delivery must be made.	- 如果恢复交付： - 如果产品交付停止后又恢复，需检查所有的交付需求，如果必要，恢复并/或重复一个R5设计评审。须作出新的M300交付决定。
Further details see [10].	进一步细节参见[10]。
The diagram illustrates the product lifecycle phases and milestones. It starts with 'Idea Collection / 想法收集' (M120), followed by 'Product Scope / 产品范围' (M150), 'Realization Concept / 实现概念' (M200), 'Implementation & Integration / 实现与集成' (M300), 'Product Verification & Validation / 产品验证和确认' (M400), 'Delivery & Product Support / 交付&产品支持' (M450), 'Discontinuance of Production / 停产' (M450), and finally 'Product Phase Out / 产品退市' (M500). Review points R1, R2, R3, R4, and R5 are marked along the timeline.	
Product phase-out (M450 – M500)	产品退市 (M450 – M500)
Start: M450 Decision	开始: M450 决定
Project request M450	项目要求M450
Purpose of the process phase	流程阶段的目的
- Phase out of a product is performed - End of product lifecycle is reached	- 产品退市实行 - 产品生命周期结束
Activities and core artifacts (that are started during this phase)	活动和核心产出物 (在此阶段开始的)
Supply of spare parts is discontinued.	停止供应备件。
As a minimum, it is mandatory to ensure that safety-related product weaknesses are rectified up to M500 (EOL).	最为最低限度，必须保证安全相关的产品缺陷在M500 (EOL) 中得到改正。
All DHF and DMR documents relating to the developed product must be kept up to date during the entire life of the product.	在整个产品生命周期里，所有研发产品相关的DHF和DMR文档必须保持最新状态。
For further details, see [10].	进一步细节参见[10]。

6 General requirements/ 通用要求

6.1 Design and development planning/ 设计和开发计划

The design and development activities are planned. The responsibilities and interfaces of the activities, which contribute to the design and development process, are documented (QMP, PMP). The plans are reviewed, updated, and approved during project development [34].	对设计和开发活动做出计划。对与设计和开发过程有关的责任和活动接口进行记录（QMP，PMP）。在项目开发过程中对计划进行审查、更新和批准 [34]。
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6.2 Tailoring/ 裁减

6.2.1 Project Specific Tailoring/ 项目特定裁减

Depending on individual project circumstances, the requirements and stipulations in this directive may be adapted in line with needs particular needs that there is appropriate justification (tailoring). However, any such changes must not compromise the basic principles of the PEP. In the case of further developments, for example, the formal M120 and M150 decisions may be omitted, providing that all the required design documents are already available. Further project-specific milestones may be necessary to ensure that the project documentation is produced in a sensible order. The reasons for the tailoring must be documented in the quality management plan.	根据具体项目情况，在由适当理由的情况下，可以依照特定需求对指令的需求和规定进行调整（裁减）。但任意此类变更均不得与 PEP 的基本原则相违背。进一步开发情况下，如所有要求的设计文档均已可用，则可省略正式 M120 和 M150 决定。可能需要针对具体项目设立节点，以确保项目文档按合理顺序生成。裁减的理由必须记录在质量管理计划中。
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6.2.2 Software Specific Tailoring/ 软件特定裁减

Additional instructions apply in the case of software development. For further details, see [39].	适用于软件开发的其它说明，更多细节参见[39]。
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6.2.3 Standard Tailoring for Successor Versions of Platform Release/ 平台发布后续版本的标准裁减

The first version of a platform is developed with all milestones (M120, R1, M150, R2, M200). Standard Tailoring: For successor projects R1/R2 and M120/M150/M200 can be combined to one R and one M milestone: R2/M200. Required milestone inputs are the sum of R1 and R2.	平台的第一个版本开发经历所有主评审（M120，R1，M150，R2，M200）。 标准裁减：对于后续项目 R1 / R2 和 M120 / M150 / M200 可以组合为一个 R 和一个 M 主评审：R2 / M200。所需的评审输入是 R1 和 R2 的总和。
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6.2.4 Standard Tailoring for Agile Approaches during Product Development/ 产品开发期间敏捷方法的标准裁减

Permissions: a. R1/R2 and M120/M150/M200 are combined to one R and one M milestone. Requirement specifications like Market	权限： a. R1/R2 和 M120/M150/M200 合并为一个 R 和一个 M 主评审。需求规范（如市场
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Requirements Specification or Engineering Requirements Specification are not expected as milestone input but planned as a committed product backlog. They will be created according to backlog planning. Details for further milestone inputs are defined in the Quality Management Plan	需求规范或工程需求规范)不是里程碑输入预期的,而是规划承诺的产品待办事项。它们将根据待办工作计划创建。质量管理计划中定义了更多评审输入的详细信息。
b. Further process results are generated step by step according to the prioritized product backlog. Meanwhile the order and the dependencies of process results must be considered. The dependencies are visible on SOP level.	进一步的过程结果将根据优先排序的产品待办事项逐步生成。同时,必须考虑过程结果的顺序和依赖关系。依赖关系在 SOP (标准操作程序) 层面上是可见的。
Constraints: a. The agile product development by itself and details about proceeding and responsibilities (e.g. sprints, iterations, done criteria, special roles) are described in the quality management plan and/or on SOP level, as well as other milestone tailoring. b. R4 and R5 are mandatory. c. All affected project members are informed.	约束: a. 敏捷产品开发本身以及有关流程和职责(例如最小迭代周期、迭代、完成标准、特殊角色)的详细信息在质量管理计划和/或 SOP 级别以及其他评审裁减中都有描述。 b. R4 和 R5 是必须的。 c. 通知所有受影响的项目成员。

6.3 Requirements for the product/ 产品需求

Requirements must be described in a structured and unambiguous manner and should contain acceptance criteria. Specifications for implementation should only be described as detailed as absolute necessary, to give the responsible competence centers or suppliers enough flexibility.	必须以结构化和明确的方式描述需求,并应包含验收标准。实施规范仅在有必要时进行详细描述,以为相关技术中心或供应商提供合理灵活性。
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6.4 Further Requirements for Development and Maintenance/开发和维护的进一步要求

6.4.2 Claims/ 声明

Claims about the product must be proven. This concerns the labelling, data sheet, instructions for use and advertising of the products, inclusive any document hand over to Siemens Healthcare external people.	必须证明有关产品的声明。这涉及产品的标签、数据表、使用说明和广告,包括移交给西门子医疗外部人员的任何文件。
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6.4.3 Product Security/ 产品网络安全

Product security aspects must be considered for product development and maintenance. Details are described in [42]	在产品开发和维护过程中,必须考虑产品网络安全方面的问题。详见[42]。
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6.4.4 AI Systems/ 人工智能系统

All AI systems developed, integrated, or placed on the market as part of the product must comply with specific regulations. Detailed implementation is governed by [41].	所有作为产品组成部分而开发、集成或投放市场的人工智能系统,均须符合特定法规。具体实施细则由[41]规定。
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6.5 Requirement tracing/ 需求追踪

The requirements from all requirement levels up to the testing levels must be demonstrably traceable through the development documents, vertically and horizontally. Trace results are documented.	所有需求层面至测试层面的需求都要在开发文档中进行垂直和水平方向的追踪。追踪结果需存档。
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6.6 Automated processes/ 自动过程

If computers or automated IT systems (referred to as "non-product software") are used in the development process, the software used must be validated to ensure it is appropriate for its intended use/intended purpose. This validation process must be completed and documented before the software is used for the first time (see [16]).	如果在开发过程中使用了计算机或者自动的 IT 系统（参见“非产品软件”），软件必须经确认其符合预期用途/预期目的。软件在第一次使用之前必须完成并用文档记录其确认过程（参见 [16]）。
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6.7 Risk management/ 风险管理

The product risk management (see [8]) is a process that is implemented over the entire life cycle of the product. The creation of the risk management plan for documenting and implementing the RM process as well as the risk assessment and documentation of the risk analysis and risk evaluation are to be started in this phase to define necessary risk mitigations to be implemented for further product development. If necessary, the risk analysis is updated, and the effectiveness of preventive measures reviewed in each process section.	产品风险管理是用于整个产品生命周期的一个过程（参见[8]）， 在这个阶段，需要创建风险管理计划，以记录和实施风险管理（RM）过程，同时进行风险评估和风险分析的风险评估和记录，以定义必要的风险缓解措施，以便在产品开发的后续阶段实施。必要的时候，风险管理和预防措施的有效性需要在每个流程中进行更新。
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6.8 Product Reliability/产品可靠性

The Product Reliability Management process shall minimize negative effect for users or customers due to limited reliability, availability, or maintainability of a product. The reliability related product requirements are handled through the development-V. For additional information see [38].	产品可靠性管理流程应尽量减少由于产品的可靠性、可用性或可维护性有限而对用户或客户的负面影响。 相关产品的可靠性要求通过开发-V 处理。 有关其他信息，请参阅 [38]。
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6.9 Product-Related Environmental Protection

The requirements regarding project-related environmental protection are aimed at the efficient implementation of product-related decisions and to mitigate concerns. Specific, measurable, and achievable environmental criteria shall be defined, that	关于项目相关环境保护的要求旨在高效执行与产品相关的决策，并减轻担忧。 应定义具体、可测量且可实现的环境标准，以支持创建环境可持续的产品。这些标准被整合到标准开发流程中。
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support the creation of environmentally sustainable products. These criteria are integrated into the standard development process.	
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6.10 Standards & Laws/ 标准和法律

Compliance with laws and standards [17] must be ensured over the entire life cycle of the product. Adjustments to the product must be assessed and measures implemented if standards or laws have been changed.	在产品的整个生命周期必须符合法律和标准[17]的要求。如果法律法规有更改，需要对产品做相应的调整评估，并实行应对措施。
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6.11 Patents and trademarks/ 专利和商标

Trademark rights must be considered: • New trademark rights respectively should be applied. • Third-party trademark rights must be considered and evaluated.	需考虑商标权： • 新的商标权可分别申请。 • 需考虑并评估第三方商标权。
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6.12 Procurement and Use of Third-Party Software/ 采购和第三方软件使用

The procedure outlined in [2] for e.g., supplier evaluation, specification, verification, and documentation must be followed for internal and external purchases of system and software components, external work, or services during the execution of the project. Third-party software is protected by copyright and offered under one or more licenses that grant the user rights while at the same time imposing obligations upon the user. Failure to comply with these license terms will constitute copyright infringement with potentially serious consequences. Therefore, measures must be taken to ensure that license obligations are fulfilled and that any additional risks resulting from the use of Third-party software are minimized.	在项目执行期间，内部和外部购买系统和软件组件、外部工作或服务时，必须遵循 [2]中概述的供应商评估、规范、验证和文档记录程序。 第三方软件受版权保护，并根据一个或多个授权许可提供给用户，这些许可在授予用户权利的同时，也对用户施加了义务。不遵守这些许可条款将构成版权侵犯，可能会带来严重的后果。因此，必须采取措施确保履行许可义务，并最小化因使用第三方软件而产生的任何额外风险。
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6.13 Preventive action/ 预防措施

Information from predecessor products must be taken into consideration when developing new products or refining existing products to systematically reduce the possibility of faults or errors. The results of appropriate investigations must be documented (error prevention report). Inputs are e.g. change request, complaints or system, design and process FMEAs.	在开发新产品或者改进现有产品的时候，必须考虑前序系统的信息，以保证系统性地减少缺陷和错误，相关的调查结果必须记录在文档中（错误预防报告）。例如，更改，抱怨，及系统、设计、流程的 FMEA。
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6.14 Concurrent engineering/ 并行开发

Concurrent engineering is permissible and means that certain sections can be processed in parallel. It is not possible to complete a design review without completing the preceding design review.	并行开发是被允许的，这意味着特定的过程可以平行进行，但是后续设计评审必须基于前面的设计评审完成。
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6.15 Transfer to Production/ 生产转移

The "transfer to production" process ensures that the development output is systematically transferred to production and/or procurement departments. The individual steps required in this process are specified in a corresponding work instruction [3].	生产转移流程保证开发的输出被系统性地转移到生产和/或采购部门。此流程的详细步骤定义在指令[3]中。
Concurrent engineering supports the transfer to production since it enables production, procurement departments (purchasing and logistics), and suppliers to be involved at an early stage.	并行开发可以更好地支持生产转移，这样生产、采购部门（采购和物流）、供应商可以更早参与项目。

6.16 Change control procedure/修改控制程序

6.16.1 Changes during product development/ 产品开发中的更改

During the product engineering process, changes to the released design input document and the resulting adaptations to design output documents must be performed in accordance with [1]. If necessary, regulations deviating from these stipulations may be set out in the quality management plan to cover the correction of documents during the development phase but prior to completion of verification. If applicable error tracking is performed according to defect management process.	在产品开发过程中，对已经发布的设计输入文档和设计输出文档结果的调整必须遵照[1]的流程。必要的时候，对这些操作的偏差需要在 QMP 中列出以覆盖在开发过程中的文档更正，但是要早于验证完成前。 如果适用，错误跟踪将根据缺陷管理流程进行。
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6.16.2 Changes after Release /Product enhancements/ 发布后的变更/产品增强

Changes to released products must be carried out in accordance with [1] and [39].	对已经发布的产品的更新必须符合[1]和[39]。
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6.17 Design review / Technical review/ 设计评审/技术评审

6.17.1 Design review/ 设计评审

Formal, documented design reviews for the development output must be planned and implemented at the end of each phase. Concurrent engineering is permissible, but it is not possible to complete a design review without completing the preceding design review. It is recommended that design reviews involve experts from affected processes to prevent errors and to ensure proper implementation.	正式，文档化的开发输出的设计评审必须提前计划并在开发的每个结束阶段实施。 允许并行工程，但如果不完成前面的设计评审，就不可能完成设计评审。 建议相关流程的专家都参与设计评审，来保证质量。 设计评审的参与、执行和记录遵照[18]和[13]或者[14]。
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Stipulations for participating in, implementing, and documenting design reviews are set out in [18] and [13] or [14]. The outcome of a design review is documented in the design history file (DHF). The product quality manager as independent person must be involved in the design review.	设计评审的输出被记录在设计历史文件 (DHF)。产品质量经理作为独立人员必须参与设计评审。
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6.17.2 Technical review / Content review/ 技术评审/内容评审

Reviews of DHF and DMR documents focus on the detailed technical content of the development work. The review of a design document is also referred to as a 'technical review' or 'content review'. For the requirements in this regard, see [18] and [13] or [14].	DHF 和 DMR 的评审关注开发工作的技术细节。对设计文档的评审可以参照“技术评审”或者“内容评审”，对于这部分的要求，请参见[18]和[13]或者[14]。
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6.18 Project phase application/ 项目阶段申请

The project phase application documents the decision, if and how the next project phase may be started. Project phase applications submitted by the product line/project management, include • cross-references to the relevant documents forming the basis of the decision, • agreement from the head of the Business line that the relevant project (sub)section can be initiated.		项目阶段申请记录决定，是否及如何开展项目下一阶段。项目阶段申请由产品线/项目管理提出，包括： • 对用于支持做决定的相关文档的交叉参考 • 业务部门领导同意项目流程（子流程）可以启动
Project Application	Decision relating to	Applicant
M120	Start of product specification	Product manager
M150	Start of system specification	Product manager
M200	Start of subsystem development	Project Lead
M300	Product release	Project Lead
M400	Product phase-out	Product manager
M450 Review	EOS decision	Product manager
项目申请	关于	申请者
M120	产品规范的开始	产品经理
M150	系统规范的开始	产品经理
M200	子系统开发的开始	项目经理
M300	产品发布	项目经理
M400	产品退市	产品经理
M450 Review	停止产品支持的决定	产品经理

6.19 Handling Deviations/偏差处理

Deviations from the prescribed process have to be evaluated – as soon as discovered and measures have to be decided at least by product quality manager and project manager. It is also recommended to involve the process manager. They are documented e.g. in design review records, change requests or defects.	规定流程中如出现偏差需立刻进行评估，应对措施至少需由产品质量经理和项目经理决定。此外，还建议邀请流程经理。其记录在例如设计审查记录，变更请求或缺陷中。
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6.20 Design verification/ 设计验证

Design and development verification shall be performed in accordance with planned and documented arrangements to ensure that the design and development outputs have met the design and development input requirements also including Hazard, S&L and Usability Requirements. Verification plans that include methods and acceptance criteria are documented.	设计和开发验证应按照计划和文件安排进行，以确保设计和开发输出满足设计和开发输入要求，包括危害、S&L 和可用性要求。记录包括方法和验收标准的验证计划。
If the intended use/intended purpose requires that the medical device be connected to, or has an interface with, other medical device(s), verification shall include confirmation that the design outputs meet design inputs when so connected or interfaced.	如果预期用途/预期目的的要求医疗设备连接到其他医疗设备或与其他医疗设备具有接口，则验证应包括确认设计输出在如此连接或接口时符合设计输入。
The verification of a medical device takes place on ZMs or their equivalents. Precondition is a successful integration of the subsystems. If the software is a subsystem, it is verified during verification of the medical device. Testing activities must ensure that every system is tested in its maximum and/or its critical configurations.	可在 ZM 或其同等设备上对医疗器械进行验证。前提是子系统的成功集成。如软件作为子系统，其验证应于医疗器械验证时进行。测试活动确保每个系统都最大限度，并/或在最关键配置环境下进行。
The requirements of the system requirement specification (including subsystem and component requirements) of the medical device as well as the concluded test specifications are available and released before performing the test cases. Test results are recorded, reviewed, documented, and archived. Tests can be supplemented / replaced by simulations, calculations, and analyses. The test records provide the evidence that the targets of the test specifications are performed, and the expected results are met. If problems or faults occur during testing, such problems or faults must be documented and analyzed; decisions must also be made regarding any necessary corrective action. Summary reports include a summary of test results and an overall evaluation of the testing phase concerned.	医疗器械的系统需求规范（包括子系统和部件规范）及总结测试规范需在执行测试前可用并发布。测试结果必须记录，审查，存档并归档在案。测试可以模拟，计算，和分析来补充或替代。测试记录提供了证据证明测试规范已达目标，预期结果已达到。如果测试中出现问题或错误，必须予以记录，分析，并作出决定，采取必要的改正措施。总结报告包括测试结果的摘要，和相关测试阶段总的评估。
The results of the design verification, including the identification of the requirements, methods, the date, and the individuals performing the verification, are documented in the design history file (DHF) [30].	设计验证的结果，包括需求的识别，方法，日期，及执行验证的人员，都记录在设计历史文件（DHF）[30]。

6.21 Design validation/ 设计确认

The validation process is carried out using series production samples to confirm that: the development result meets the product requirements and fulfills the intended	确认流程以一系列生产样例来实现，以确认： - 开发结果满足产品需求和预期用途/预期目的， - 服务和装配/安装文档是准确的（仅当有硬件
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use/intended purpose, - the service and assembly/installation documentation are accurate (only if hardware is developed); - product features in documents published for customers (e.g. technical data, product development documents) are accurate; - Accompanying documents (see [12]) are suitable for the intended use/intended purpose of the product; - the functional design can be reproduced.	开发时)； - 发布给客户的文档中的产品特性（如技术数据，产品开发文档）是准确的 - 附属文档（见[12]）对于预期的产品用途/预期目的是合适的。 - 功能设计可以重现。
Design and development validation shall be performed in accordance with planned and documented arrangements to ensure that the resulting product can meet the requirements for the specified application or intended use/intended purpose also including hazard, S&L and usability requirements. Validation plans that include methods and acceptance criteria are documented. Both during the formative and the summative validation phase the users of the medical device are involved, depending on its complexity: the validation is performed under simulated conditions (iCUT) or under real conditions directly at user sites (CUT). At least one of them is mandatory. The summative validation of the medical device is performed on FM's (initial production samples, lots, or batches, or their equivalents).	设计和开发确认应按照计划和文件安排进行，以确保最终产品能够满足指定应用或预期用途/预期目的的要求，包括危害、S&L 和可用性要求。记录包含方法和验收标准的验证计划。 根据医疗器械的复杂性，在形成性和总结性确认期间，用户都需参与其中：在模拟条件（iCUT）下，或直接在用户端实际条件（CUT）下进行确认。其中之一是必须的。 医疗器械的总结性确认应在FM生产样机（初始生产样品，份额，批，或它们的等价物）上进行。
The product requirements of the medical device as well as the concluded test specifications are available and released before performing the test cases. Where samples are used a rationale for sample sizes shall be provided based on statistical techniques. The confidence interval is defined and justified. If the intended use/intended purpose requires that the medical device be connected to, or has an interface with, other medical device(s), validation shall include confirmation that the requirements for the specified application or intended use/intended purpose have been met when so connected or interfaced. The evidence of the design validation is given by the records from the user-oriented tests, which are summarized in a report with an overall evaluation. If acceptance criteria are not achieved during validation, such problems must be documented and analyzed; decisions must also be made regarding any necessary corrective action.	医疗器械的产品需求以及总结测试规范需在执行测试前可用并发布。 使用样本时，应根据统计技术提供样本大小的合理数。置信区间已定义并对齐。 如果预期用途/预期目的要求医疗器械连接到其他医疗器械或与其他医疗器械有接口，则确认应包括确认要求对于指定的应用或预期用途/预期目的，在如此连接或接口时已得到满足。 设计确认的证据是由面向用户的测试给出，总结评估并记录在报告中。如果确认过程中没有达到验收标准，这些问题必须予以记录和分析；并对任何必要的纠正措施作出决定。

The results of the design validation, including identification of the design, methods, the date, and the individuals performing the validation, are documented in the design history file [28].	设计确认的结果, 包括设计的识别, 方法, 日期, 及执行确认的人员, 都记录在设计历史文件 [28]。
If drawing samples have been demonstrably reworked based on approved documents from production, they can be classified as production samples in terms of their technical status and can be used for the purposes of design validation.	如果图纸样例已经基于核准的生产文档明确返工, 可以根据他们的技术状态进行生产样例分类, 并可以用于设计确认。
Validation shall be completed prior to release for use of the product to the customer.	设计确认应在向客户发布产品之前完成。

6.22 Process validation/ 过程确认

Transfer to production must be completed before process validation can begin. Series production samples, but no reworked drawing samples, are acceptable to use as evidence for a process validation. Process validation is described in [4].	在流程确认开始前, 生产转移必须先完成。系列产品样例, 但不是重新翻新的图纸样机, 可以用作过程确认的证据。过程确认过程在[4]中描述。
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6.23 Clinical evaluation/ 临床评估

Clinical evaluation undertaken for the development of a medical device: Premarket research and development are guided by clinical evaluation and risk management. Typically, manufacturers carry out clinical evaluations to • define needs regarding clinical safety and clinical performance of the device; • in case of possible equivalence to an existing device, evaluate if there are clinical data available and determine equivalence; • carry out a gap analysis and define which data still need to be generated with the device under evaluation, whether clinical investigations are necessary and if so, to define the study design; As the initial clinical evaluation identifies the questions to be answered by a clinical investigation, the clinical evaluation process should generally commence in advance of any clinical investigation.	开展医疗器械的临床评估: 进入市场前的研究与开发要以临床评估和风险管理为指导。 通常, 制造商用临床评估来: • 定义设备临床安全性和临床性能的需求; • 如果可能与现有设备相当, 评估是否有临床数据可用并确定等效性; • 进行差距分析, 确定被评估的设备仍然需要生成哪些数据, 是否需要临床调查, 如果是, 定义研究设计; 如果初步临床评估确定了临床调查要回答的问题, 临床评估过程一般应在任何临床调查之前开始。
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6.24 Software Development/软件开发

Software as medical device and as subsystem of a medical device is defined and created according to [35], [39] and [41].	软件作为医疗设备和医疗设备的子系统, 根据 [35], [39]和[41]定义并创建。
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6.25 OTS Software/ 现成软件

Third-Party Software is protected by copyright and offered under one or more licenses that grant	第三方软件受版权保护, 并根据一个或多个许可证提供, 这些许可证授予用户权利, 同时对用户
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the user rights while at the same time imposing obligations upon the user. Failure to comply with these license terms will constitute copyright infringement with potentially serious consequences. Therefore, measures must be taken to ensure that license obligations are fulfilled and that any additional risks resulting from the use of Third-Party Software are minimized.	施加义务。 不遵守这些许可条款将构成侵犯版权，并可能产生严重后果。因此，必须采取措施确保履行许可义务，并将因使用第三方软件而导致的任何额外风险降至最低。
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6.26 Unique Device Identification/ 唯一的设备识别

The serialization representative creates a unique device identifier (UDI) for each medical device and every device package. The uniqueness is accomplished by using GTIN's according to GS1 standard [36], [37].	序列化代表为每个医疗设备和每个设备包创建唯一的设备标别（UDI）。其唯一性是通过使用根据GS1标准的GTIN实现的 [36], [37]。
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7 Document/ 文档

7.1 Document management/ 文档管理

The project manager allocates responsibility for developing and maintaining the MDF/DHF to the document manager. During his/her activities, the document manager must ensure at all times that:	项目经理将 MDF/DHF 的开发和维护任务分配给文档经理。作为他/她的任务，文档经理必须始终确保：
- document plans are developed and suitably structured. - the necessary documents for a process section are properly planned. - planned access rights are set up for the completed documentation. - document changes are properly planned	- 开发清晰合理结构的文档计划 - 计划必要的流程文档 - 对整个文档计划访问权限 - 计划文档的更改
The stipulations set out in [7] apply. Document planning at the start of a process section must be subject to a technical review before approval and implementation. The use of document plans for each design review is governed by the rules set out in [13] or [14].	文档管理的条款请参照[7]。过程开始阶段的文档计划在正式被批准和实施前必须进行技术评审。每个设计评审中如何使用这个文档计划请参照[13]或[14]中的规定。

7.2 Quality management plan (QMP)/ 质量管理计划（QMP）

The quality management plan is a summary of the general quality-related instructions and documents an essential part of the design planning, such as: Project organization, including the functional areas, tailoring of the development process and responsibilities, as well as specifications	质量管理计划是质量相关的指导总结，并且记录了设计规划中的一个重要部分，例如： - 项目组织，包括功能领域，开发流程裁减和责任，以及对其他组织单位的接口 - 适用的评审方法，包括在必须的评审中应该参与的人员
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for interfaces to other organizational units • Review methods to be applied, including persons to be involved in the mandatory reviews • Testing concept • Quality goals • Criteria for terminating and repeating process sections • Quality management of products and services provided by third parties. • Reasons for, and description of procedures deviating from procedures of the present regulations (e.g. tailoring) • Corrective actions arising from the error prevention report. Procedure for ensuring the acquisition of country-specific approvals.	• 测试总纲 • 质量目标 • 需要终止和重复某个过程的标准 • 第三方提供的产品和服务的质量管理 • 与现行法规的程序中描述的开发流程的偏差（裁减）和偏差的原因 • 包括从错误预防报告而来的纠正措施，确保获得国家特定批准的程序。
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7.3 Project management plan (PMP) / 项目管理计划（PMP）

The project management plan is based on the quality management plan and describes the project at the current point in time in detail. It includes the following, for example: • The product structure (as a precursor to a system flowchart) • The project structure plan, with the responsibilities for the scope of goods and ◦ services in the product, with its subsystems, system components and modules/software components • Specifications for product-specific documents (e.g. for production, testing, user information. etc.) • Work output required for creating the documents (e.g. drafts, interim results, plans, investigation results, etc.) • Aids required for product manufacture (e.g. testing stations, production tools, etc.) • Time Schedule • Work package structure plan, as a personnel/resources plan • Description of the organizational interfaces, deadlines, and cost apportionment if several Healthcare organizational units are involved • Project risks and mitigations Note: The QMP and PMP may be combined in one document.	此项目管理计划基于质量管理计划，详细描述了在当前节点下的项目，包括： • 产品结构（作为系统流程图的前身） • 项目结构计划，包括产品（以及子系统，系统组件，模块/软件组件等）和服务的范围 • 特定产品的需求说明书(如用于生产，测试，和用户信息等) • 用于创建文档所需的工作输出(如草稿，中间结果，计划，调研结果等) • 生产制造所需的辅助(如测试站，生产工具等) • 时间计划 • 工作包结构计划，作为员工/资源计划 • 组织接口描述，最后期限，成本分配，如果涉及数个 Healthcare 组织部门 • 项目风险和风险管理 注意：QMP 和 PMP 可以结合于一个文档中。
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7.4 Product requirements specification level/ 产品需求规范层级

The product specification is part of the design input documents. It describes the required product features in a complete and detailed manner so that they can later be used as the basis for product development as well as for design validation. The product specification contains the product features from the user's (intended use) and stakeholder's (e.g. SCM and CS) perspective and is typically generated at system level. The product requirements specification level describes the required product functions as requirement to the product or its use: Use specification: • Intended use/intended purpose and indications for use including usability requirements • Intended user and patients • Use scenarios with use requirements Product specification: In BL CT, the product requirement specification is called market requirement specification (MRS). • Target markets • Technical requirements to the system or subsystems • Product quality requirements • Requirements to deliverables (enhancements, accessories, instructions for use) • Requirements for product reliability as well as expected service life. • Specification of necessary measures to implement the after-sales service strategy (process requirements). • Specification of necessary measures to implement the production and logistics concept (process requirements). • If applicable: Sustainability requirements.	产品规范是设计输入文档的一部分。 它以完整和详细的方式描述所需的产品特性，以便它们稍后可以作为产品开发和设计验证的基础。 产品规范包含了从用户（预期用途）和利益相关者（例如供应链管理SCM和客户服务CS）的角度出发的产品特性，并且通常在系统级别生成。 产品需求规范层级描述了所需的产品功能或用途： 使用规范： • 预期用途/预期目的及适用性，包括可用性需求 • 预期用户和患者 • 使用场景及使用需求 产品规范：在CT业务线，产品需求规范被称为市场需求规范（MRS） • 目标市场 • 系统或子系统的技术要求 • 产品质量要求 • 发运需求（增强，附件，使用说明） • 产品可靠性的要求以及预期的使用寿命。 • 实行售后服务策略的必要措施规范（流程需求） • 实行生产和物流概念必要措施的规范（流程需求） • 可选：首个发运日（上市时间），市场定价（目标成本），质量目标。 • 如适用：可持续性要求。
The objective is, to collect enough information for planning and realization of user research activities. Based on user research knowledge the product requirements specification is redefined during project runtime.	其目标是收集足够信息以规划和实现用户研究活动。 基于用户研究，产品的需求规范可在项目运行时重新定义。

7.5 System requirements specification level/ 系统需求说明级别

The product requirements are translated into technical product concepts and system requirements. The system requirement specification describes the requested technical	产品需求转化为技术产品概念和系统需求。系统需求规范描述了要求的技术性能及相应的质量属性（性能和非功能性需求，如可维护性，操
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functions and their corresponding quality attributes (performance and nonfunctional requirements e.g. maintainability, interoperability, usability). The focus is to describe the external system interfaces to/from the user and to/from the environment (black box). The content is based on the scope and intended use/intended purpose of the product. The key elements of a system requirements specification are: User interface specification: • The user interface specification describes comprehensively and prospectively the user interface and considers the risk analysis and the use specification. The user interface specification should be developed ahead of the user interface and includes testable technical requirements for the user interface. • Note: In projects where the document structure restarts based on anterior conditions (e.g. projects from the continuous product line) the user interface design may be part of different specifications created in underneath requirement levels. • Indicates whether accompanying documentation and training are required. System (functions) requirement specification: • System architecture (decomposition of the product into subsystems by evaluating the system requirements and with focus on internal interfaces) • Functional structure • Measures arising from the error prevention report • Description of the external interfaces • Applicable standards and laws requirements for country-specific approvals, including the definition of measures to fulfill those • requirements for user documentation	作性，可用性)。重点描述与用户和外界（黑盒子）的外部系统接口。其内容基于产品的范围及预期用途/预期目的。系统需求说明书包括如下关键因素： 用户界面规范： • 用户界面规范全面和前瞻性地描述了用户界面，并考虑其风险分析和使用说明书。用户界面规范应早于用户界面的开发，并包括所述用户界面可测试的技术要求。 • 注意：当项目的文档结构是基于先前条件重启（例如项目是基于之前的产品线），用户界面设计可以是另外的创建于下层需求的一部分。 • 指出是否需要随机文档和培训。 系统（功能）需求规范： • 系统架构（评估系统需求，将产品分解至子系统，重点描述内部接口） • 功能结构 • 由预防错误报告产生的措施 • 外部接口的描述 • 适用于特定国家批准的标准和法律要求，包括满足这些要求的措施的定义 • 用户文档的要求
Before the system requirements specification is prepared, technical product concepts must be selected and evaluated, taking the following factors into consideration: • First delivery date (time to market) • Market pricing (target costing) • Quality target	系统需求说明准备好之前，必须考虑下列因素，选择并评估技术产品概念： • 首个交货期（上市时间） • 市场定价（成本目标） • 质量目标
As defined in [33], system requirements include for software:	如[33]所定义，系统需求包括软件相关的： • 功能和性能需求

- Functional and capability requirements - Software system inputs and outputs - Interfaces between the software system and other systems as well as internal interfaces - Software-driven alarms, warnings, and operator messages - Information security requirements - User interface requirements implemented by software (usability engineering requirements) - Data definition and database requirements - Installation and acceptance requirements of the delivered medical device software at the operation and maintenance site or sites - Requirements related to methods of operation and maintenance - Requirements related to IT-network aspects - User maintenance requirements and - Regulatory requirements	- 软件系统输入和输出 - 软件系统与其他系统以及内部接口之间的接口 - 软件驱动的报警，警告和操作人员信息 - 信息安全需求 - 软件实现的用户界面需求（可用性工程需求） - 数据定义和数据库需求 - 交付的在操作站点和维护站点的医疗设备软件的安装和验收需求 - 与操作和维护方法相关的需求 - 与IT网络相关的需求 - 用户维护需求和 - 法规需求
In addition, - Risk control measures for potential hardware and software faults/errors - Requirements for the software architecture (as separate documents, if appropriate) - Testability requirements	以及： - 潜在的硬件和软件故障/失效的风险控制措施 - 软件架构的需求（如果恰当，作为单独的文档） - 可测试性需求
The specification also includes requirements for the OTS software to be used: - Functional and performance requirements for the OTS software to be used; - System hardware and software requirements to support proper operation.	该规范还包括要使用的OTS软件需要满足： - 要使用的OTS软件的功能和性能需求； - 支持正确操作的系统硬件和软件需求。
For software as a medical device (SaMD), the system requirements are derived from the product requirements.	对于作为医疗设备（SaMD）的软件，其需求来源于产品需求。
For embedded software (i.e. software as a subsystem), the software(sub)system requirements are defined by refining the requirements from the system level.	对于嵌入式软件（即作为子系统的软件），软件（子）系统需求需要通过从系统级别需求来提炼定义。

7.6 Subsystem / component level/ 子系统/部件级别

For all subsystems defined by system architecture a requirement specification is created. The content of these documents is derived from the system requirements specification level, it	- 创建由系统架构中定义的所有子系统的需求规范。这些文档的内容源于系统需求规范，着重于内部接口和子系统功能的描述。 - 创建子系统或软件架构。
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focuses on internal interfaces and describes the functions of the subsystem. • A subsystem or software architecture is created. • For all components defined by subsystem / software architecture a requirement specification is created. The content of these documents is derived from the subsystem specification level, it focuses on internal interfaces and describes the functions of the component. • Design specifications contains detailed descriptions of the hardware and/or software component (including algorithms). Design specifications are necessary when there are complex functions that must be implemented. If function and interfaces are clear from a drawing, circuit diagram or e.g. UML diagram without additional information, this drawing diagram can be used as a design document.	• 创建由子系统/软件架构中定义的所有组件的需求规范。这些文档的内容源于子系统需求规范，着重于内部接口和组件功能的描述。 • 设计规范包含对硬件和软件组件（包括算法）的详细描述。当需要实现复杂功能时，设计规范是必需的。如果图纸，电路图（如 UML 示意图）上的功能和接口很清晰，不需额外信息加以补充，可以作为设计文档。
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7.7 System Realization concept/ 系统实施概念

Description of the concept for system architecture, system interfaces, and typical product features from a system perspective.	基于系统角度的系统架构概念，系统接口，和典型产品特性的描述
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7.8 Test concept/ 测试概念

The test concept contains information about • the test setup on the respective testing levels to be applied, • test scope, methods, responsibilities, and timeline of planned tests, • test start/end criteria. Where samples are used a rational for sample sizes shall be provided based on statistical techniques according to [21]. The confidence interval is defined and justified. It can be described in a dedicated document or be part of the quality management plan.	测试概念包含 • 在各自的测试级别上应用的测试设置 • 计划测试的测试范围、方法、责任和时间表 • 测试开始/结束标准 在使用样本的地方，应根据统计技术提供样本大小的合理理由，参照[21]。置信区间被定义并得到论证。 可以在专门的文档中描述，也可以作为质量管理计划的一部分。
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7.9 Results from risk management/ 来自于风险管理的结果

Product specific risk management results are described [8].	[8]中描述了特定产品的风险管理结果。
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7.10 Error prevention report/错误预防报告

Report on preventive measures based on data from predecessor products. See 6.13.	基于前身产品数据采取的防范措施报告。参见 6.13。
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7.11 Document plan / design history file (DHF) / 文档计划/设计历史文件（DHF）

Document Plan/DHF shall be created and maintained for every product. The DHF is included in Medical Device File (MDF) The Design History File is a compilation of records describing the development history of a finished device. It contains or references the records necessary to demonstrate that the design was developed in accordance with the approved design plan and the requirements of this section [31].	每个产品都需要创建并维护相应的文档计划/设计历史文件。 设计历史文件是包含在医疗器械文档中（MDF）。设计历史文件是一系列描述完成设备开发历史的记录。它包括或涉及必要的记录，用于表明设计是按照批准的设计计划和此章节[31]需求开发的。
Document plans: • in certain circumstances, refer to the document plans of subsystems, system components, modules and parts, including bought-in items, which together form the product; • refer to all documents and records that contain the engineering history, the manufacturing processes, and the specifications for a finished device; • are created for each product variant and are maintained throughout the entire product life cycle. For further rules relating to document management, see [7].	文档计划: • 在某些条件下，参考子系统，系统组件，模块和部件，包括组成产品的买入部件的文档计划； • 涉及包括成品开发历史，生产流程和规格的所有文档及记录； • 每个产品均需创建文档计划并在整个产品生命周期中进行维护； 文档管理的详细规定参考[7]。

7.12 Device history record (DHR) / 产品追溯性记录（DHR）

The device history record is a compilation of records describing the production and service history of a finished device. It ensures for each batch, lot, or unit that proofs are maintained to demonstrate that the device is manufactured in accordance with the DMR and the requirements of this section. For further and detailed information of DHR see [15],[32] The DHR includes or refer to the following information:	产品追溯性记录是一系列用于描述完成产品生产和服务历史的记录。它确保每一批，每一份，每一单元制造出来的产品，都有据可查，是符合产品制造性文档，和此章节要求的。 产品追溯性记录的更详细信息参考[15], [32] 产品追溯性记录（DHR）包含或参考以下信息：
• Date of manufacture • Quantity manufactured • Quantity released for distribution • Acceptance records which demonstrate the device is manufactured in accordance with the DMR, • Parameters of special processes, • Primary identification label and labeling used for each production unit • Any device identifications and control numbers used • final product release	• 生产日期 • 制造数量 • 发放分发的数量 • 验收记录，证明设备是按照 DMR 制造的 • 特殊过程的参数 • 主要识别标签和用于每个生产单元的标签 • 使用的任何设备标识和控制编号 • 最终产品放行

7.13 Device master record (DMR)/ 产品制造性文档（DMR）

The Device Master Record is a compilation of records describing the production processes and specifications for the finished device. The device master record for each product type and production site includes or refers to the following information: • Device specifications including appropriate drawings, composition, formulation, component specifications and software specifications • Production process specifications including the appropriate equipment specifications, production methods, production procedures and production environment specifications • Quality assurance procedures and specifications including acceptance criteria and the quality assurance equipment to be used • Packaging and labeling specifications, including methods and processes used Installation, maintenance and servicing procedures and methods. [33]	产品制造性文档是一系列用于描述生产流程和完成设备说明的记录。每种产品和生产站点的产品制造性文档都包括或涉及下列信息： • 设备说明，包括合适的图纸，构图，公式，组件说明书和软件说明 • 生产流程说明，包括合适的设备说明，生产方法，生产流程和生产环境说明 • 质量保证流程和说明，包括接受标准和将使用的质量保证设备 • 打包和标签说明，包括使用的方法和流程；安装，维护和服务流程及方法。[33]
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7.14 Medical Device File (MDF)/ 医疗器械文档（MDF）

For Each medical device type or medical device family a file/files shall be established and maintained either containing or referencing documents generated to demonstrate conformity with the requirements of ISO 13485:2016 and compliance with applicable regulatory requirements. • General description of the medical device, intended use/intended purpose, and labelling, including any instructions for use, • Specifications of product, • Usability engineering file, • Specifications or procedures for manufacturing, packaging, storage, handling and distribution, • Procedures for measuring and monitoring, • As appropriate, requirements for installation and • As appropriate, procedures for servicing. The medical device file (MDF) consists of the design history file (DHF) and the device master record (DMR).	对于每个医疗设备型号或医疗设备系列，应该建立和维护一套文件，可以是参考文档，用来证明设备符合 ISO13485：2016 要求并且遵守适当的法规要求。该文件的内容应包括但不限于： • 医疗器械的一般描述，用途，以及标签，包括任何使用说明， • 产品规格， • 可用性工程文件， • 制造，包装，储存，处理和分发的规范或程序， • 测量和监测程序， • 视情况而定，安装要求和 • 维修程序。 医疗器械文档（MDF）由设计历史文件（DHF）和产品制造性文档（DMR）组成。
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7.15 Technical documentation/ 技术文档

The technical documentation is an organized, clearly presentable, unambiguous, and readily searchable documentation according to the Medical Device Regulation 2017/745 according to Annex II, Medical Devices Directive 93/42/EEC and other Harmonized Union Legislation which apply to the device in question (e.g. RoHS, Machinery Directive). The technical documentation comprises the device description and its specification, reference to previous and similar devices, information provided with the product, design stages and manufacturing process including suppliers and subcontractors, demonstration of conformity, benefit-risk analysis, verification and validation, pre-clinical and clinical data.	根据医疗器械法规2017/745附件II，医疗器械指令93/42 / EEC和其他适用于相关设备的协调联盟立法（例如RoHS，机械指令），技术文档是有组织的，清晰呈现的，明确且易于搜索的文档。技术文档包括设备描述及其规格、对以前和类似设备的引用、随产品提供的信息、设计阶段和制造过程（包括供应商和分包商）、一致性证明、收益风险分析、验证和确认、临床前和临床数据。
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7.16 Country-specific approvals (regulatory approval plan) / 特定国家核准（法规核准计划）

Following M150, the project manager and the regulatory affair team define the work packages, evidence, and documentation required in order to obtain the country-specific approval and summarize these requirements in a regulatory approval plan. Applications for approval are submitted by quality management department. The product place on market in China shall comply with the requirement of [40].	根据M150，项目经理和法规部门定义工作包，证明，所需文档，以获得特定国家的批准，并在法规核准计划中总结这些需求。 核准申请由质量管理部门递交。 产品在中国上市应遵守[40]的要求。
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7.17 Affidavit for Service Software/ 服务软件宣誓书

The affidavit confirms that Siemens Healthcare is the sole owner of the intellectual property in the service software concerned and thus owns the marketing rights. The affidavit is signed by the Head of Business Process (R&D). The original of the affidavit is sent to the marketing department of the unit Customer Services and is archived in the DHF.	宣誓书确认西门子医疗是相关服务软件知识产权的唯一所有者，因此拥有营销权。宣誓书由业务流程（R&D）负责人签署。宣誓书的原件被发送到客户服务部门的营销部门，并存档在 DHF 中。
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7.18 Usability engineering file/ 可用性工程文件

The usability engineering file consists of all the documents and records that prove the appropriate usability of the product to be developed. For details see [19].	可用性工程文件由所有用于证明将要开发的产品合适可用性的文档和记录组成。详情参阅[19]。
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7.19 System flowchart/ 系统流程图

Description of the possible combinations of hardware and/or software configurations.	关于软硬件配置的可能组合描述
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7.20 Accompanying documents/ 随机文档

Documents for the operator and end user as specified in [12] to ensure that a device can be used in a safe and proper manner according to its intended use/intended purpose. Any residual risks, contra-indications and any undesirable side-effects, including information to be conveyed to the patient must be described in the instructions for use documents. Documents for customer service to ensure that installation, service support, and maintenance are properly carried out.	[12]里详述的，操作者和终端用户文档是用于确保设备按照预期用途/预期目的在安全和适当的方式下使用的。任何残留风险，禁忌症和任何不良副作用，包括要传达给患者的信息必须在使用说明文件中描述。 文档被用于客户服务的安装，服务支持，及实施维护的文档。
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7.21 Financial product plan (FPP) / 财务产品计划

The financial product plan is required for: • M150 • M200 • M300	财务产品计划对下列对象是必需的： • M150 • M200 • M300
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8 Tasks and responsibilities/ 任务和职责

8.1 Management of SSME CT / SSME CT 管理层

Management of SSME CT • Initiates the creation of a project milestone approval in which all processes are represented. • Ensures that the project-specific tasks, responsibility, and authority of the project manager are determined and documented. • Signs the CE declaration of conformity.	SSME CT 管理层 • 启动创建一个包含所有流程的项目里程碑审批。 • 确保确定并写下项目相关工作、项目经理的职责和权限。 • 在 CE 符合声明上签字
Project Milestone Approval (PMA) • Tracks and evaluates the progress of the project based on economic risks (e.g. it assesses the financial effects of canceling the project, delaying it or increasing resources used.) • Makes decisions on project phase applications based on the documents submitted.	项目里程碑批准 (PMA) • 跟踪并基于经济风险对项目进展进行评估 (例如，评估取消项目、延期或追加资源所带来的财务影响)。 • 基于提交的文档对项目阶段申请做决定。 • 正式批准每个项目阶段申请，并签字。这个签字意味着对要求的资源认可 (财

Formally approves each project phase application by signing the application concerned. This signature authorizes release of the requested resources (capital investment, personnel, and financial resources).	务投资、人员、财务的资源)。
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8.2 Quality Management Representative / 质量管理负责人

confirms on project phase applications that the project plans are consistent with the quality management system currently in force.	对于项目阶段申请，确认质量计划与生效的质量管理体系的一致性
ensures that all standards and laws currently in force and applicable to the project are made available.	确保项目遵守所有生效的法律和法规
assigns an independent SSME CT Quality Control Engineer for each project/product	确保为每一个项目和产品有一个独立的质量控制工程师。
signs the CE Declaration of Conformity	在 CE 符合声明上签字

8.3 Product Manager / 产品经理

Initiates the products defined in the roadmaps in a timely and proper manner and prepare business plans.	及时发起路线图中定义的产品，并准备业务计划。
Defines product requirements and prepares product requirements specifications.	定义产品需求，并准备产品需求文件。
Ensures that requirements are implemented in a customer-oriented manner in close cooperation with the project management.	与项目管理紧密合作，确保需求以客户为导向的方式在项目中实现。
Coordinates limiting conditions for a product phase-out.	协调产品逐步退市的预值条件。

8.4 Project Manager / 项目经理

- Is responsible for design and development planning and tracking. - Appoints the PSG members in consultation with their respective functional departments (definition and appointment in the PMP, QMP, or PQMP),	- 负责设计和开发计划和跟踪 - 与有关职能部门协商，任命项目管理小组成员（项目管理计划，质量管理计划，或 PQMP 的定义和任命）
Leads and motivates the product core team (members are product manager, project engineer, marketing manager, business administration).	领导和激励产品的核心团队（成员包括产品经理，项目工程师，营销经理，工商管理）。
Leads the PSG and acts as the representative for the overall project	带领项目管理小组，并且是整个项目的代表人

management,	
Represents the interests of those functional departments not represented in the PSG,	承担代表未参加项目管理小组的那些职能部门利益的责任
ensures that, the PSG is made up of experts from the relevant functional departments, e.g. marketing, product definition, development, production, logistics, customer services, commercial department, quality management, and procurement, depending on the process and tasks involved,	根据被牵涉的流程和任务，确保项目管理小组由各职能部门的专家组成，如：市场部，产品定义，研发部，生产部，物流部，客户服务部，商务部，质量部，采购部。
Plans, manages, and controls the project, including, for example:	对项目进行计划、管理和控制，包括：
Preparing and updating the QMP/PMP as well as document plans with support of document manager.	- 制定和更新质量管理计划和项目管理计划 - 以及在文档经理协助下完成文档计划。
- Ensuring the project documentation is in place. - Performing design reviews	- 确保项目文档的生成 - 执行设计评审
- Supporting patents application. - Ensuring that the necessary spare parts are planned in sufficient time. - Assuming responsibility for compliance with process requirements, for the creation and maintenance of the documentation in the DHF and DMR, and for the overall product quality achieved.	- 支持申请专利 - 确保有充足的时间计划准备必要的备件 - 承担符合流程要求，开发历史文档（DHF）和产品制造性文档（DMR）的创建和维护，整个产品质量达标的责任

8.5 Product Quality Manager / 产品质量经理

- carries out checks on the project from an independent perspective and is an independent participant of design reviews. - Must stop processes and projects if the relevant quality requirements are not fulfilled (e.g. in terms of product features, intended use/intended purpose, or compliance with QM system). - Escalates any relevant issues directly to the quality management representative of the SSME CT management. - Creates summary technical documentation based on product documentation.	- 从独立的角度对项目进行检查，并且是设计评审的独立参与者。 - 如果未满足相关质量要求（例如，在产品功能、预期用途/预期目的或符合 QM 体系方面），则必须停止流程和项目。 - 将任何相关问题直接上报给 SSME CT 管理层的质量管理代表。 - 根据产品文档创建汇总技术文档。
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8.6 Product Steering Group (PSG)/ 项目管理小组成员

• Supports the project manager. • Remains responsible for the product even after the start of delivery, although the composition of the group can change. • Makes project decisions – one representative each affected functional area is essential. • Decides about any changes to milestones, changed work packages endangering the project timeline or phases, any incidents affecting the project or in case of occurrence of project risks • Decides whether milestones must be repeated when changes are made to product scope. • Decides on any action to be taken as a result of the outcome from a design review	• 支持项目经理 • 尽管团队组成在产品交付后可能变化，仍对产品负责 • 做出项目决策 – 每个受影响的功能领域都必须有一名代表。 • 决定里程碑的任何更改，危及项目时间表或阶段的工作包更改，影响项目的任何事件或发生项目风险的情况 • 决定在更改产品范围时是否必须重复里程碑。 • 决定设计评审后采取的任何行动
Every member of the PSG • Is a delegate representing the respective functional area, • Coordinates tasks between the PSG and the respective functional area, • Makes project-relevant decisions for the respective functional area. • Ensures that the regulations applicable to the project are implemented. • Escalates topics to PSG which cannot be solved within the respective functional area, and upward along the hierarchical structures if it cannot be solved there. The tasks and responsibilities of the PSG members are specified in detail in the QMP.	每个项目管理小组的成员： • 是各个职能领域的代表， • 协调功能区域和 PSG 之间的任务， • 为其所在的功能部门作与项目有关的决定。 • 确保项目相关的规定得到实施。 • 将不能在各功能区域内解决的问题汇报到 PSG，如仍未解决，汇报到更上层。 项目管理小组成员的任务和职责详细描述于质量管理计划（QMP）中。

8.7 Responsibilities during delivery / 发运中的职责

The following processes cover the most important activities during product support: • Supply chain management (SCM): Production (procurement), delivery (incl. DHR preparation), installation and commissioning, improvements to the production process	以下是在产品支持过程中会参与的主要流程： • 供应链管理（SCM）：生产（采购），发运（包括 DHR 准备），安装和投产，生产过程的提高
• Product lifecycle management (PLM): Error correction, product upgrades	• 产品生命周期管理（PLM）：错误纠正，产品升级。

Quality management: Monitoring and evaluating product quality	质量管理：监控和评估产品的质量
- Customer service (CS): - Provision of repair and maintenance including supply of spare parts - Provision of update / upgrade instructions - Escalation of service problems to specific persons within CT development and to CT management.	- 客户服务（CS）： - 提供维修和维护服务包括备件 - 提供更新和升级的指导 - 将服务过程中的问题汇报给 CT 指定的开发人员以及管理人员

9 Provisional solution and backward method/ 过渡措施和补救办法

These instructions are applied to new projects on or after the effective date. For on-going projects before the design review R5, the QMP must be supplemented specifying how and when a transition will be made to the PEP. The transition must be completed 6 months after the validity date of these instructions at the latest. Process phases which have already been concluded through the design review do not need to be revised.	本规定适用于处于生效日期或在此之后的新项目。对于正在进行的、主评审 R5 之前的项目，必须在质量管理计划中补充说明如何以及何时进行 PEP 的过渡。最晚必须在本指令生效后的 6 个月内完成过渡。对于已经通过主评审而结束的过程阶段不必再进行修改。
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10 Record/记录

Record no. 记录编号	Record name 记录名称	Where stored. 保存地	Retention 保存期限
n.a.			

11 Appendix/附录

Appendix no. 附录号	Appendix name 附录名称
1	Example of using the V-model in sub-systems 在子系统中使用 V 字型模式的实例
2	Example of "Project Phase Application". "项目阶段申请"
3	Example of "Transfer Protocol" "转移报告"实例
4	Example of "Production Review Protocol" "生产评审报告"实例
5	Example of DMR/DHF document index DMR/DHF 文档索引的实例

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